

CHAPTER X

ORTHOGONAL POLYNOMIALS

The standard textbook on this subject is the book by Szegö (1939) to which we shall refer frequently. There is a systematic bibliography up to 1938, by Shohat, Hille, and Walsh (1940). Although the present chapter is concerned with orthogonal *polynomials* only, in the introductory sections we consider more generally systems of orthogonal functions. For further information on this latter topic the reader may be referred to books by Kaczmarz and Steinhaus (1935) and by Tricomi (1948), and by Vitali and Sansone (1946).

10.1. Systems of orthogonal functions

With an interval (a, b) and a *weight function* $w(x)$ which is *non-negative* there, we may associate the scalar product

$$(1) \quad (\varphi_1, \varphi_2) \equiv \int_a^b w(x) \varphi_1(x) \varphi_2(x) dx$$

which is defined for all functions φ for which $w^{1/2} \varphi$ is quadratically integrable in (a, b) . More generally, a scalar product may be defined by a Stieltjes integral

$$(2) \quad (\varphi_1, \varphi_2) \equiv \int_a^b \varphi_1(x) \varphi_2(x) d\alpha(x)$$

where $\alpha(x)$ is a non-decreasing function. If $\alpha(x)$ is absolutely continuous, (2) reduces to (1) with $w(x) = \alpha'(x)$. On the other hand, if $\alpha(x)$ is a jump function, that is constant except for jumps of the magnitude w_i at $x = x_i$, then (2) reduces to a sum

$$(3) \quad (\varphi_1, \varphi_2) = \sum_i w_i \varphi_1(x_i) \varphi_2(x_i)$$

which is the appropriate definition for functions of a discrete variable.

The above definitions refer to real functions of a real variable, and to this case we shall restrict ourselves throughout this chapter. If the functions in question are complex-valued, or else if the domain of inte-

gration is an arc in the complex plane rather than a segment of the real axis, then $\varphi_2(x)$ in all these definitions must be replaced by the conjugate complex quantity.

Except in the last few sections (where we use definition (3)), we shall use definition (1) mostly, and shall assume moreover that $w(x)$ is positive almost everywhere and integrable. It should be mentioned, however, that many of the results of the introductory sections hold for the definition (2), and therefore also for the definition (3), of a scalar product.

Two functions are said to be *orthogonal* if their scalar product vanishes. A family of functions is an *orthogonal system*, on the interval (a, b) and with the weight function $w(x)$ (or distribution $a(x)$), if for any two *distinct* members of the family, $(\varphi_1, \varphi_2) = 0$. Since the space of quadratically integrable functions is *separable*, it follows that an orthogonal system consists either of a finite number or at most of a denumerable infinity of elements. Thus an orthogonal system can always be written as a (finite or infinite) *sequence*, $\varphi_0(x), \varphi_1(x), \dots$ or shortly $\{\varphi_n(x)\}$, and the orthogonal property is then expressed as

$$(4) \quad (\varphi_h, \varphi_k) = 0 \quad h \neq k.$$

We shall assume that $\{\varphi_n(x)\}$ does not contain any null function, i.e., that (φ_h, φ_h) is positive for all h . It is then easy to see that the functions of any finite subset of an orthogonal system are linearly independent, that is that a relation of the form

$$(5) \quad c_0 \varphi_0(x) + c_1 \varphi_1(x) + \dots + c_k \varphi_k(x) = 0$$

cannot be valid almost everywhere in (a, b) , except when $c_0 = c_1 = \dots = c_k = 0$. (Form the scalar product with $\varphi_h(x)$ for $h = 0, 1, \dots, k$.)

The functions $\{\varphi_n(x)\}$ form an *orthonormal system* if

$$(6) \quad (\varphi_h, \varphi_k) = \begin{cases} 0 & \text{if } h \neq k, \\ 1 & \text{if } h = k. \end{cases}$$

Every orthogonal system can be *normalized* by replacing $\varphi_h(x)$ by

$$(\varphi_h, \varphi_h)^{-1/2} \varphi_h(x).$$

A (finite or infinite) sequence $\{\psi_n(x)\}$ of linearly independent functions can be *orthogonalized* with respect to the scalar product (2) by the formation of suitable linear combinations. For instance we may put recurrently

$$(7) \quad \varphi_0(x) = \psi_0(x)$$

$$\varphi_1(x) = \mu_{10} \varphi_0(x) + \psi_1(x)$$

$$\dots \quad \dots \quad \dots \quad \dots$$

$$\varphi_n(x) = \mu_{n0} \varphi_0(x) + \mu_{n1} \varphi_1(x) + \dots + \mu_{n, n-1} \varphi_{n-1}(x) + \psi_n(x)$$

and see that $\{\varphi_n(x)\}$ is an orthogonal system if we take

$$(8) \quad \mu_{nm} = -(\psi_n, \varphi_m)/(\varphi_m, \varphi_m) \quad m = 0, 1, \dots, n-1.$$

Alternatively, we may put

$$(9) \quad \varphi_n(x) = \lambda_{n0} \psi_0(x) + \lambda_{n1} \psi_1(x) + \dots + \lambda_{nn} \psi_n(x) \quad \lambda_{nn} \neq 0$$

and determine the λ 's so that $\{\varphi_n(x)\}$ is an orthogonal system. One possible determination leads to

$$(10) \quad \phi_n(x) = \begin{array}{cccc} (\psi_0, \psi_0) & (\psi_0, \psi_1) & \cdots & (\psi_0, \psi_n) \\ (\psi_1, \psi_0) & (\psi_1, \psi_1) & \cdots & (\psi_1, \psi_n) \\ \vdots & \vdots & \ddots & \vdots \\ (\psi_{n-1}, \psi_0) & (\psi_{n-1}, \psi_1) & \cdots & (\psi_{n-1}, \psi_n) \end{array}$$

$$\begin{array}{ccc} \psi_0(x) & \psi_1(x) & \psi_n(x) \end{array}$$

It is clear that $\{\phi_n(x)\}$ is an orthogonal system, for (10) is orthogonal to $\psi_0(x), \psi_1(x), \dots, \psi_{n-1}(x)$ and hence to $\phi_m(x)$ for all $m < n$. Moreover, any orthogonal system of the form (9) is a constant multiple of $\{\phi_n(x)\}$.

In order to normalize the system (9), we introduce *Gram's determinant* G_n which is the cofactor of $\psi_{n+1}(x)$ in the expression (10) for $\phi_{n+1}(x)$. G_n is also the discriminant of the positive definite quadratic form

$$\int_a^b [\xi_0 \psi_0(x) + \dots + \xi_n \psi_n(x)]^2 w(x) dx$$

in $\xi_0, \dots, \xi_n$, and hence positive. We also put $G_{-1} = 1$. The orthonormal system of the form (9) with $\lambda_{nn} > 0$ is then uniquely determined as

$$(11) \quad \varphi_n(x) = (G_{n-1} G_n)^{-1/2} \phi_n(x).$$

Furthermore the following integral representation can be established

$$(12) \quad \phi_n(x) = [(n-1)!]^{-1} \int^{(n)} \Psi_{n-1}(\xi_0, \dots, \xi_{n-1}) \Psi_n(\xi_0, \dots, \xi_{n-1}, x) \\ \times w(\xi_0) \cdots w(\xi_{n-1}) d\xi_0 \cdots d\xi_{n-1} \quad n = 1, 2, \dots$$

where the integral is an n -tuple integral over (a, b) and

$$(13) \quad \Psi_n(x_0, \dots, x_n) = \begin{vmatrix} \psi_0(x_0) & \psi_1(x_0) & \cdots & \psi_n(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_0(x_n) & \psi_1(x_n) & \cdots & \psi_n(x_n) \end{vmatrix}.$$

(See Szegő, 1939, sec. 2.1.)

In this chapter we shall be concerned with the orthogonalization, in the form (9), of the functions $\psi_n(x) = x^n$. Thus we obtain a sequence of

orthogonal polynomials $\{p_n(x)\}, n = 0, 1, 2, \dots$ where $p_k(x)$ is a polynomial in x of exact degree k , and $(p_h, p_k) = 0$ for $h, k = 0, 1, 2, \dots$ and $h \neq k$.

The interval and the weight function (or distribution) determine the system of orthogonal polynomials up to an arbitrary constant factor in each $p_n(x)$. The polynomials may be standardized by the adoption of additional requirements. The three most frequently used additional requirements are: (i) $\{p_n(x)\}$ shall be an orthonormal system and the coefficient of x^n in $p_n(x)$ shall be positive; (ii) the coefficient of x^n in $p_n(x)$ shall have a prescribed value (usually unity); (iii) for a given x_0 (for instance $x_0 = a$), $p_n(x_0)$ shall have a prescribed value.

10.2. The approximation problem

Let L_w^2 be the class of all functions $f(x)$ for which the (Lebesgue) integral

$$\int_a^b w(x) [f(x)]^2 dx$$

exists and is finite, and let $\{\varphi_n(x)\}$ be an orthonormal system in L_w^2 . In approximating any function $f(x)$ of L_w^2 by a linear combination

$$c_0 \varphi_0(x) + \dots + c_n \varphi_n(x),$$

we regard

$$(1) \quad I_n(c_h) = \int_a^b w(x) [f(x) - c_0 \varphi_0(x) - \dots - c_n \varphi_n(x)]^2 dx$$

as the measure of accuracy of this approximation. It is easy to see that the best possible choice for c_h is that of the Fourier coefficients

$$(2) \quad a_h = (f, \varphi_h).$$

In fact, expanding $[\dots]^2$ in (1) we find

$$\begin{aligned} I_n(c_h) &= \int_a^b w(x) [f(x)]^2 dx + \sum_{h=0}^n c_h^2 - 2 \sum_{h=0}^n a_h c_h \\ &= \int_a^b w(x) [f(x)]^2 dx - \sum_{h=0}^n a_h^2 + \sum_{h=0}^n (c_h - a_h)^2, \end{aligned}$$

that is the best approximation is the $(n+1)$ st partial sum of the (generalized) *Fourier series*

$$(3) \quad a_0 \varphi_0(x) + a_1 \varphi_1(x) + \dots$$

of $f(x)$, and the measure of the accuracy of this approximation is

$$(4) \quad I_n(a_h) = \int_a^b w(x) [f(x)]^2 dx - \sum_{h=0}^n a_h^2.$$

Since $I_n(a_h) \geq 0$, it follows that $\sum a_h^2$ is convergent and we have *Bessel's inequality*

$$(5) \quad \sum_{h=0}^{\infty} a_h^2 \leq \int_a^b w(x) [f(x)]^2 dx.$$

It may happen that *Parseval's formula*

$$(6) \quad \sum_{h=0}^{\infty} a_h^2 = \int_a^b w(x) [f(x)]^2 dx$$

holds for every function $f(x)$ of L_w^2 . Then the orthonormal system $\{\varphi_n(x)\}$ is said to be *closed* in L_w^2 . In this case clearly

$$(7) \quad \int_a^b w(x) [f(x) - \sum_{h=1}^n a_h \varphi_h(x)]^2 dx \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and we say that the partial sums of the Fourier series (3) *converge in the mean* to $f(x)$. In L_w^2 , every closed orthogonal system is also *complete*, i.e., if $(f, \varphi_h) = 0$ for all h , then $f(x)$ vanishes almost everywhere. This is a consequence of the Riesz-Fischer theorem (cf. for instance Kaczmarz and Steinhaus, 1935, or Tricomi, 1948, sec. 3.3).

For a finite interval (a, b) every function of L_w^2 can be approximated arbitrarily closely, in the mean, by a continuous function, and by the theorem of Weierstrass the continuous function can be approximated by a polynomial. Thus for a finite interval and $\psi_n(x) = x^n$, or $\varphi_n(x) = p_n(x)$, we may make $I_n(a_h)$ arbitrarily small by making n sufficiently large. In other words, any system of orthogonal polynomials for a *finite interval* is *closed*. This need no longer be true if the interval (a, b) is of infinite length (Szegő 1939, sec. 3.1).

10.3. General properties of orthogonal polynomials

A weight function $w(x)$ on an interval (a, b) determines a system of orthogonal polynomials $\{p_n(x)\}$ uniquely apart from a constant factor in each polynomial. The numbers

$$(1) \quad c_n = \int_a^b w(x) x^n dx$$

are the *moments* of the weight function, and with $\psi_n(x) = x^n$ we have

$$(2) \quad (\psi_m, \psi_n) = c_{m+n}.$$

In the notation of sec. 10.1 we then have

$$(3) \quad G_n = \begin{vmatrix} c_0 & c_1 & \cdots & c_n \\ c_1 & c_2 & \cdots & c_{n+1} \\ \cdots & \cdots & \cdots & \cdots \\ c_n & c_{n+1} & \cdots & c_{2n} \end{vmatrix}, \quad \Psi_n = \begin{vmatrix} 1 & x_0 & \cdots & x_0^n \\ 1 & x_1 & \cdots & x_1^n \\ \cdots & \cdots & \cdots & \cdots \\ 1 & x_n & \cdots & x_n^n \end{vmatrix} = \prod_{r>s} (x_r - x_s).$$

If the (undetermined) coefficient of x^n in $p_n(x)$ is denoted by k_n , we have

$$(4) \quad p_n(x) = \frac{k_n}{G_{n-1}} \begin{vmatrix} c_0 & c_1 & \cdots & c_n \\ c_1 & c_2 & \cdots & c_{n+1} \\ \cdots & \cdots & \cdots & \cdots \\ c_{n-1} & c_n & \cdots & c_{2n-1} \\ 1 & x & \cdots & x^n \end{vmatrix}$$

$$(5) \quad p_n(x) = \frac{k_n}{n! G_{n-1}} \int^{(n)} \prod_{r>s} (\xi_r - \xi_s)^2 \prod_{\nu=1}^n [(x - \xi_\nu) w(\xi_\nu) d\xi_\nu]$$

Since $1, x, \dots, x^{n-1}$ are orthogonal to $p_n(x)$, we have

$$(6) \quad h_n = (p_n, p_n) = k_n^2 \frac{G_n}{G_{n-1}}.$$

For the *normalized* polynomials $k_n = (G_{n-1}/G_n)^{1/2}$, but we shall not standardize our polynomials at this stage.

Any polynomial of degree $m < n$ is a linear combination of $p_0(x), p_1(x), \dots, p_m(x)$ and hence orthogonal to $p_n(x)$. This leads to a simple proof of the following theorem on the zeros of orthogonal polynomials. *All zeros of $p_n(x)$ are simple, and located in the interior of the interval (a, b) . For if $p_n(x)$ changed its sign in (a, b) only at $m < n$ points, we could construct a polynomial $\pi_m(x)$ of degree m so that $p_n(x) \pi_m(x) \geq 0$ in (a, b) , and this contradicts $(p_n, \pi_m) = 0$. It can also be shown that between two consecutive zeros of $p_n(x)$ there is exactly one zero of $p_{n+1}(x)$, and at least one zero of $p_m(x)$ for each $m > n$ (Szegő, 1939, sec. 3.3).*

Any three consecutive polynomials are connected by a linear relation. We use the following notations: k_n is the coefficient of x^n , and k'_n the coefficient of x^{n-1} , in $p_n(x)$; $r_n = k'_n/k_n$, and $h_n = (p_n, p_n)$. We shall then prove the *recurrence formula*

$$(7) \quad p_{n+1}(x) = (A_n x + B_n) p_n(x) - C_n p_{n-1}(x) \quad n = 1, 2, 3, \dots$$

in which

$$(8) \quad A_n = k_{n+1}/k_n, \quad B_n = A_n(r_{n+1} - r_n),$$

$$C_n = A_n h_n / (A_{n-1} h_{n-1}) = k_{n+1} k_{n-1} h_n / (k_n^2 h_{n-1}).$$

To prove (7), we remark that with the value (8) of A_n , the expression $p_{n+1}(x) - A_n x p_n(x)$ is a polynomial of degree n or less, and consequently of the form

$$\gamma_0 p_n(x) + \gamma_1 p_{n-1}(x) + \cdots + \gamma_n p_0(x).$$

From the orthogonal property of the $p_n(x)$, we find that $\gamma_2 = \gamma_3 = \cdots = \gamma_n = 0$, and

$$-A_n(p_n, xp_{n-1}) = \gamma_1(p_{n-1}, p_{n-1}).$$

Now, $x p_{n-1}(x) - (k_{n-1}/k_n) p_n(x)$ is a polynomial of degree $n-1$ or less, and hence

$$-A_n h_n k_{n-1}/h_n = \gamma_1 h_{n-1}$$

or $\gamma_1 = C_n$. Lastly, the value of B_n follows on comparing coefficients of x^n on both sides of (7). The recurrence formula (7) remains valid for $n=0$ if we put

$$(9) \quad p_{-1}(x) = 0.$$

This convention will be retained throughout this chapter.

It may be noted that conversely, a system of polynomials satisfying a recurrence relation (7) with positive A_n and C_n , is an orthogonal system.

From (7) we easily obtain the *Christoffel-Darboux formula*

$$(10) \quad \sum_{\nu=0}^n h_{\nu}^{-1} p_{\nu}(x) p_{\nu}(y) = \frac{k_n}{k_{n+1} h_n} \frac{p_{n+1}(x) p_n(y) - p_n(x) p_{n+1}(y)}{x - y}$$

and for $y \rightarrow x$,

$$(11) \quad \sum_{\nu=0}^n h_{\nu}^{-1} [p_{\nu}(x)]^2 = \frac{k_n}{k_{n+1} h_n} [p_n(x) p'_{n+1}(x) - p'_n(x) p_{n+1}(x)].$$

Let $\{p_n(x)\}$ be the system of orthogonal polynomials for the weight function $w(x)$, and let $\rho(x)$ be a polynomial of degree l which is non-negative in (a, b) and has simple zeros at $x_1, x_2, \dots, x_l$. The orthogonal polynomials $q_n(x)$, belonging to the weight function $\rho(x) w(x)$ are then given by *Christoffel's formula*

$$(12) \quad c_n \rho(x) q_n(x) = \begin{array}{cccc} p_n(x) & p_{n+1}(x) & \cdots & p_{n+l}(x) \\ p_n(x_1) & p_{n+1}(x_1) & \cdots & p_{n+l}(x_1) \\ \dots & \dots & \dots & \dots \\ p_n(x_l) & p_{n+1}(x_l) & \cdots & p_{n+l}(x_l) \end{array}$$

in which c_n is an arbitrary constant factor (Szegő, 1939, sec. 2.5). If some of the zeros of $\rho(x)$ are multiple zeros, (12) must be replaced by a confluent form.

Orthogonal polynomials have some important extremum properties. The first of these can be derived from the result at the beginning of sec. 10.2 and reads: *The integral*

$$(13) \quad \int_a^b |\pi_n(x)|^2 w(x) dx$$

in which $\pi_n(x)$ denotes any polynomial of degree n with the leading term x^n becomes a minimum if and only if $\pi_n(x) = \epsilon k_n^{-1} p_n(x)$ where ϵ is a constant and $|\epsilon| = 1$. The second property involves the polynomials

$$(14) \quad K_n(x, y) = \sum_{m=1}^n h_m^{-1} p_m(\bar{x}) p_m(y)$$

which are defined for complex x, y ($\bar{x}$ is the conjugate complex of x). We may remark here that for finite x_0, a and for $x_0 \leq a$, the polynomials $K_n(x_0, x)$ are orthogonal with respect to the weight function $(x - x_0)w(x)$ (cf. (10) and (11)). The extremum property in question may be formulated as follows (Szegő, 1939, theorem 3.1.3). *Let $\pi_n(x)$ be an arbitrary polynomial of degree n with complex coefficients such that the integral (13) is equal to unity. For any fixed (possibly complex) x_0 the maximum of $|\pi_n(x_0)|^2$ is reached if and only if*

$$\pi_n(x) = \epsilon [K_n(x_0, x_0)]^{-1/2} K_n(x_0, x)$$

where $|\epsilon| = 1$. The maximum itself is $K_n(x_0, x_0)$.

10.4. Mechanical quadrature

Many interesting properties of orthogonal polynomials depend on their connection with problems of interpolation and mechanical quadrature. In this section we can give no more than a brief description of some of the basic results, and refer to Szegő's book (1939, sec. 3.4, chapters XIV, XV) for further information.

Let $x_1, x_2, \dots, x_n$ be n distinct points of the interval (a, b) and let

$$(1) \quad \pi_n(x) = (x - x_1)(x - x_2) \cdots (x - x_n),$$

$$l_{\nu}(x) = (x - x_{\nu})^{-1} \pi_n(x) / \pi_n'(x_{\nu}) \quad \nu = 1, \dots, n.$$

The $l_{\nu}(x)$ are the fundamental polynomials associated with the abscissae $x_1, \dots, x_n$ in the Lagrangean interpolation

$$(2) \quad L(x) = \sum_{\nu=1}^n f(x_{\nu}) l_{\nu}(x)$$

of the function $f(x)$.

If the integral

$$(3) \quad I = \int_a^b w(x) f(x) dx$$

is to be computed for a function whose values at the x_{ν} are given, it seems natural to use (2) and compute

$$(4) \quad J = \int_a^b w(x) L(x) dx = \sum_{\nu=1}^n f(x_{\nu}) \int_a^b w(x) l_{\nu}(x) dx$$

in the expectation that J will be an approximation to I . Actually, for any $x_1, \dots, x_n$, we have $I = J$ for all polynomials $f(x)$ of degree $\leq n-1$. However, if we choose the x_{ν} to be the n zeros of $p_n(x)$, the orthogonal polynomial of degree n associated with the weight function $w(x)$, then $I = J$ for all polynomials $f(x)$ of degree $\leq 2n-1$. For in this case $f(x) - L(x)$ is a polynomial of degree $\leq 2n-1$ vanishing at all the zeros of $p_n(x)$ and hence of the form $p_n(x) \pi_{n-1}(x)$ where $\pi_{n-1}(x)$ is a polynomial of degree $\leq n-1$. Then

$$I - J = \int_a^b w(x) [f(x) - L(x)] dx = (p_n, \pi_{n-1}) = 0.$$

It is customary to write

$$(5) \quad J = \int_a^b w(x) L(x) dx = \sum_{\nu=1}^n \lambda_{\nu n} f(x_{\nu})$$

where the $\lambda_{\nu n}$ are called the *Christoffel numbers*. They are connected with the moments of $w(x)$ by the relations

$$(6) \quad \sum_{\nu=1}^n x_{\nu}^h \lambda_{\nu n} = c_h \quad h = 0, 1, \dots, n-1$$

obtained by choosing $f(x) = x^h$. The Christoffel numbers are positive, and the following formulas hold:

$$(7) \quad \lambda_{\nu n} = \int_a^b \frac{w(x) p_n(x)}{p_n'(x_{\nu})(x - x_{\nu})} dx = \int_a^b w(x) \left[\frac{p_n(x)}{p_n'(x_{\nu})(x - x_{\nu})} \right]^2 dx$$

$$(8) \quad \lambda_{\nu} = \frac{k_{n+1} h_n / k_n}{p'_n(x_{\nu}) p_{n+1}(x_{\nu})} = \frac{1}{K(x_{\nu}, x_{\nu})}$$

If we denote by $x_{1n}, x_{2n}, \dots, x_{nn}$ the n zeros of $p_n(x)$ and by $y_{1n}, \dots, y_{nn}$ the n numbers in (a, b) defined by

$$(9) \quad \int_{-}^{y_{\nu n}} w(x) dx = \lambda_{1n} + \dots + \lambda_{\nu n} = \Lambda_{\nu n}$$

then we have a number of *separation theorems*

$$(10) \quad x_{\nu-1, n} < x_{\nu, n+1} < x_{\nu, n}$$

$$(11) \quad y_{\nu-1, n} < y_{\nu, n+1} < y_{\nu, n}$$

$$(12) \quad x_{\nu, n} < y_{\nu, n} < x_{\nu+1, n}$$

$$(13) \quad \Lambda_{\nu-1, n} < \Lambda_{\nu, n+1} < \Lambda_{\nu, n}.$$

10.5. Continued fractions

The recurrence formula 10.3(7) suggests the *continued fraction*

$$(1) \quad \frac{1}{|A_0 x + B_0|} - \frac{C_1}{|A_1 x + B_1|} - \frac{C_2}{|A_2 x + B_2|} - \dots,$$

where A_n, B_n, C_n are given by 10.3(8). The n^{th} convergent R_n/S_n is defined as the finite fraction obtained by stopping at the term $A_{n-1}x + B_{n-1}$ in (1) so that

$$(2) \quad R_0 = 0, \quad S_0 = 1; \quad R_1 = 1, \quad S_1 = A_0 x + B_0 = p_1(x)/p_0(x).$$

Both R_n and S_n satisfy the recurrence relation

$$(3) \quad X_{n+1} = (A_n x + B_n) X_n - C_n X_{n-1}.$$

The initial conditions are

$$(4) \quad \text{for } R_n: X_0 = 0, X_1 = 1; \quad \text{for } S_n: X_0 = 1, X_1 = p_1(x)/p_0(x).$$

Referring to 10.3(7) it is seen that

$$(5) \quad S_n = p_n(x)/p_0(x) = k_0^{-1} p_n(x).$$

In order to express also R_n , we introduce the associated polynomial

$$(6) \quad q_n(x) = \int_a^b \frac{p_n(x) - p_n(t)}{x - t} w(t) dt$$

which is a polynomial of degree $n - 1$. From 10.3 (7)

$$\begin{aligned} q_{n+1}(x) - (A_n x + B_n) q_n(x) + C_n q_{n-1}(x) \\ = -A_n \int_a^b p_n(t) w(t) dt = 0 \end{aligned} \quad n = 1, 2, \dots$$

Moreover, $q_0(x) = 0$, $q_1(x) = \int_a^b k_1 w(t) dt = k_1 c_0$, and hence

$$(7) \quad R_n = (k_1 c_0)^{-1} q_n(x).$$

We thus see that R_n/S_n is a rational function of x with simple poles at $x = x_{\nu n}$. The residues at these poles can be computed from

$$\lim_{x \rightarrow x_{\nu n}} (x - x_{\nu n}) \frac{q_n(x)}{p_n(x)} = \frac{1}{p'_n(x_{\nu n})} \int_a^b \frac{p_n(t)}{t - x_{\nu n}} w(t) dt = \lambda_{\nu n},$$

see 10.4 (7), and we have the decomposition in partial fractions

$$(8) \quad \frac{R_n}{S_n} = \frac{k_0}{k_1 c_0} \sum_{\nu=1}^n \frac{\lambda_{\nu n}}{x - x_{\nu n}}.$$

On expansion of the sum in descending powers of x the relation 10.4 (6) shows that the first $2n$ coefficients are the moments c_h . Hence we obtain formally

$$(9) \quad \lim_{n \rightarrow \infty} \frac{R_n}{S_n} = \frac{k_0}{k_1 c_0} \sum_{h=0}^{\infty} \frac{c_h}{x^{h+1}}.$$

For a finite interval (a, b) , and for any x in the complex plane cut along the segment (a, b) of the real axis, Markoff proved that $\lim R_n/S_n$ exists and (9) is valid. Moreover,

$$(10) \quad \lim_{n \rightarrow \infty} \frac{R_n}{S_n} = \frac{k_0}{k_1 c_0} \int_a^b \frac{w(t)}{x - t} dt$$

in this case (Szegő, 1939, sec. 3.5). Intervals of infinite length present formidable difficulties which are discussed in the theory of (Stieltjes and Hamburger) *moment problems*. For these see Shohat and Tamarkin (1943).

10.6. The classical polynomials

The orthogonal polynomials belonging to the intervals and weight functions listed in the following table arise very frequently and have

been studied in great detail. They are known as the *classical orthogonal polynomials*.

CLASSICAL ORTHOGONAL POLYNOMIALS

a	b	$w(x)$	NAME
-1	1	1	Legendre or spherical
-1	1	$(1-x^2)^{\lambda-1/2}$	Gegenbauer or ultraspherical
-1	1	$(1-x)^\alpha(1+x)^\beta$	Jacobi or hypergeometric
$-\infty$	∞	$\exp(-x^2)$	Hermite
0	∞	$x^\alpha e^{-x}$	(generalized) Laguerre.

All these polynomials have a number of properties in common of which the three most important ones are:

- (i) $\{p_n'(x)\}$ is a system of orthogonal polynomials;
- (ii) $p_n(x)$ satisfies a differential equation of the form

$$A(x)y'' + B(x)y' + \lambda_n y = 0$$

where $A(x)$ and $B(x)$ are independent of n , and λ_n is independent of x ;

- (iii) there is a generalized Rodrigues' formula

$$(1) \quad p_n(x) = \frac{1}{K_n w(x)} \frac{d^n}{dx^n} [w(x) X^n]$$

where K_n is a constant and X is a polynomial in x whose coefficients are independent of n .

Conversely, any of these three properties characterizes the classical orthogonal polynomials in the sense that any system of orthogonal polynomials which has one of these properties can be reduced to a classical system. For (i), this has been proved by Hahn (1935) and Krall (1936); for (ii) by Bochner (1939) (in this case there are some trivial exceptions); and for (iii) by Tricomi (1948a). We shall briefly indicate the argument in this last case.

Let $\{p_n(x)\}$ be a sequence of polynomials, $p_n(x)$ of exact degree n , for which (1) holds for every $n = 0, 1, 2, \dots$, the polynomial X being of degree k . Note that it is not necessary to assume that the $p_n(x)$ are orthogonal polynomials or that $w(x)$ is a weight function. From (1), with $n = 1$, we have

$$(2) \quad K_1 p_1(x) = X' + X w'(x)/w(x).$$

First let $k = 0$. Then X is a constant and w'/w is a linear function of x . By a linear change of the independent variable we may make $w'/w = -2x$, hence $w = \exp(-x^2)$, and the polynomials are the Hermite polynomials, see 10.13(7). Next let $k = 1$. Then a linear change of x brings

$$(3) \quad \frac{w'(x)}{w(x)} = \frac{K_1 p_1(x) - X'}{X}$$

into the form $w'/w = -1 + \alpha/x$, so that $X = x$, $w = x^\alpha e^{-x}$, and we have the Laguerre polynomials, see 10.12(5).

We now discuss $k \geq 2$. In this case we may take

$$(4) \quad X = \prod_{r=1}^k (x - a_r)$$

and at first we assume all the a_r different from each other. From (3)

$$\frac{w'(x)}{w(x)} = \sum_{r=1}^k \frac{a_r}{x - a_r}$$

so that (1) becomes

$$p_n(x) = K_n \prod_{r=1}^k (x - a_r)^{-a_r} \frac{d^n}{dx^n} \left[\prod_{r=1}^k (x - a_r)^{n+a_r} \right],$$

and for $n = 2$ this fails to be a quadratic polynomial except when $k = 2$. The case of repeated factors in (4) can be excluded by a similar consideration, so that in (4) we must have $k = 2$, $a_1 \neq a_2$. By a linear change of x we may make $a_1 = -1$, $a_2 = 1$, and write

$$X = (1 - x)^2, \quad w(x) = (1 - x)^\alpha (1 + x)^\beta$$

so that this case leads to Jacobi polynomials, see 10.8(10).

It may be mentioned that Hahn (1949) has extended these results considerably. He replaced the differential operator $df(x)/dx$ by the more general linear operator

$$Lf(x) = \frac{f(qx + \omega) - f(x)}{(q - 1)x + \omega}$$

and showed that in this more general case each of conditions (i), (ii), (iii), and of two further conditions, characterizes the same family of orthogonal polynomials. The classical polynomials are limiting cases of Hahn's polynomials, and so are the polynomials of sections 22-25.

10.7. General properties of the classical orthogonal polynomials

Many important properties of the classical orthogonal polynomials follow easily from the generalized Rodrigues' formula 10.6(1). We assume $\alpha > -1$ in the Laguerre case and $\alpha > -1, \beta > -1$ in the Jacobi case.

In each case we have in 10.6(1) a $w(x)$ which is non-negative and integrable in (a, b) . Moreover, since all derivatives up to and including the $(n-1)$ st of $w(x) X^n$ vanish at a and b , we may integrate by parts n times in

$$(f, p_n) = K_n^{-1} \int_a^b f(x) \frac{d^n}{dx^n} [w(x) X^n] dx,$$

obtain

$$(f, p_n) = (-1)^n K_n^{-1} \int_a^b f^{(n)}(x) w(x) X^n dx$$

and hence $(f, p_n) = 0$ if f is a polynomial of degree $< n$. In other words, the polynomials 10.6(1) form an orthogonal system in the interval (a, b) with the weight function $w(x)$, and all the results of the previous sections are valid for these functions. In particular, we have the recurrence formula 10.3(7) with the notation 10.3(8) which we shall use again in the present section.

In deriving the differential equation from 10.6(1) we shall write D instead of d/dx . From 10.6(1) and from Leibniz' formula of the differentiation of a product we have

$$\begin{aligned} D^{n+1} [XD(wX^n)] &= K_n [XD^2(wp_n) + (n+1)X'D(wp_n) \\ &\quad + \frac{1}{2}n(n+1)X''wp_n]. \end{aligned}$$

On the other hand, using 10.6(3),

$$\begin{aligned} D^{n+1} [XD(wX^n)] &= D^{n+1} \{ [K_1 p_1 + (n-1)X'] wX^n \} \\ &= K_n \{ [K_1 p_1 + (n-1)X'] D(wp_n) + (n+1)[K_1 p_1' + (n-1)X''] wp_n \} \end{aligned}$$

since $K_1 p_1 + (n-1)X'$ is at most a linear function of x . Comparison of the two results yields the differential equation

$$(1) \quad X \frac{d^2 y}{dx^2} + K_1 p_1(x) \frac{dy}{dx} + \lambda_n y = 0$$

for $y = p_n(x)$, where

$$(2) \quad \lambda_n = -n [k_1 K_1 + \frac{1}{2}(n-1)X''].$$

The self-adjoint form of the differential equation is

$$(3) \quad \frac{d}{dx} \left[X w(x) \frac{dy}{dx} \right] + \lambda_n w(x) y = 0.$$

For the details of the proof see Tricomi (1948a, p. 210-212). Since X is at most a quadratic polynomial, and $p_1(x)$ is a linear polynomial, the differential equation (1) can be reduced to the hypergeometric equation or to one of its special or limiting cases.

For the classical polynomials we also have the *differentiation formula*

$$(4) \quad X \frac{dp_n(x)}{dx} = (a_n + \frac{1}{2} n X'' x) p_n(x) + \beta_n p_{n-1}(x)$$

where

$$(5) \quad a_n = n X'(0) - \frac{1}{2} X'' r_n, \quad A_n \beta_n = -C_n [k_1 K_1 + (n - \frac{1}{2}) X''],$$

and A_n, C_n, k_n, r_n have the same meaning as in sec. 10.3. By means of 10.3(7), the right-hand side of (4) can be expressed in terms of p_n and p_{n+1} .

The proof of (4) in Tricomi (1948a, p. 212-215) is based on the fact that

$$X p_n'(x) - \frac{1}{2} n X'' x p_n(x)$$

is a polynomial of degree $\leq n$ and hence of the form

$$a_n p_n(x) + \beta_n p_{n-1}(x) + \gamma_2 p_{n-2}(x) + \dots + \gamma_n p_0(x).$$

The coefficients $a_n, \dots, \gamma_n$ are then determined by the orthogonal property. In the determination of β_n the differential equation (3) is also used.

Finally we note that by n successive integrations by parts as at the beginning of this section,

$$(6) \quad h_n = (p_n, p_n) = (-1)^n k_n n! K_n^{-1} \int_a^b X^n w(x) dx,$$

from 10.4(8), 10.3(7), and (4)

$$(7) \quad \begin{aligned} \lambda_{\nu, n} &= A_{n-1} h_{n-1} [X(x_{\nu, n})/\beta_n] [p_{n-1}(x_{\nu, n})]^{-2} \\ &= A_{n-1} h_{n-1} [\beta_n/X(x_{\nu, n})] [p_n'(x_{\nu, n})]^{-2}, \end{aligned}$$

and from (6)

$$(8) \quad (-1)^n k_n K_n > 0.$$

Each of the following six sections is devoted to one of the principal families of classical orthogonal polynomials. Each of these six sections is organized on the following plan:

- (i) Standardization of the polynomials.
 - (ii) Computation of the ten constants
- (9) $h_n, k_n, r_n, A_n, B_n, C_n, K_n, \lambda_n, a_n, \beta_n$
 given by 10.7(6), 10.3(8), 10.7(2), 10.7(5).
- (iii) Statement of the recurrence relation, differential equation, and other relations, except that whenever these relations are cumbersome, it will be left to the reader to substitute the values of the ten constants (9) into the general formulas of this and the previous sections.
 - (iv) Connection with functions of the hypergeometric type and complete integration of the differential equation.
 - (v) Generating function or functions.
 - (vi) Integral representations.
 - (vii) Addition theorems, series expansions, and miscellaneous results.
- Asymptotic properties, zeros, expansion problems will be discussed in later sections.

We shall use the notation

$$(10) \quad D = \frac{d}{dx}$$

and shall put

$$(11) \quad (a)_0 = 1, \quad (a)_n = \frac{\Gamma(a+n)}{\Gamma(a)} = a(a+1) \cdots (a+n-1).$$

Accounts of the classical orthogonal polynomials are given in the works referred to in the introduction, and also in the book by Magnus and Oberhettinger (1948, Chap. V).

10.8. Jacobi polynomials

We shall use Szegő's notation $P_n^{(\alpha, \beta)}(x)$ for the suitably standardized orthogonal polynomials associated with

$$(1) \quad a = -1, \quad b = 1, \quad w(x) = (1-x)^\alpha(1+x)^\beta, \quad X = 1 - x^2.$$

In order to make the weight function non-negative and integrable, we assume

$$(2) \quad a > -1, \quad \beta > -1.$$

Many of the formal relations are valid without this restriction.

(i) *Standardization.*

$$(3) \quad P_n^{(\alpha, \beta)}(1) = \binom{n+\alpha}{n} = \frac{(\alpha+1)_n}{n!}.$$

(ii) *Constants.*

$$(4) \quad (2n + \alpha + \beta + 1) n! \Gamma(n + \alpha + \beta + 1) h_n = 2^{\alpha+\beta+1} \Gamma(n + \alpha + 1) \Gamma(n + \beta + 1)$$

$$(5) \quad k_n = 2^{-n} \binom{2n + \alpha + \beta}{n}, \quad r_n = \frac{n(\alpha - \beta)}{2n + \alpha + \beta}$$

$$(6) \quad 2(n+1)(n + \alpha + \beta + 1) A_n = (2n + \alpha + \beta + 1)(2n + \alpha + \beta + 2)$$

$$(7) \quad 2(n+1)(n + \alpha + \beta + 1)(2n + \alpha + \beta) B_n = (\alpha^2 - \beta^2)(2n + \alpha + \beta + 1)$$

$$(8) \quad (n+1)(n + \alpha + \beta + 1)(2n + \alpha + \beta) C_n = (n + \alpha)(n + \beta)(2n + \alpha + \beta + 2)$$

$$(9) \quad K_n = (-2)^n n!, \quad \lambda_n = n(n + \alpha + \beta + 1), \quad \alpha_n = r_n,$$

$$(2n + \alpha + \beta) \beta_n = 2(n + \alpha)(n + \beta)$$

(iii) *Rodrigues' formula.*

$$(10) \quad 2^n n! P_n^{(\alpha, \beta)}(x) = (-1)^n (1-x)^{-\alpha} (1+x)^{-\beta} D^n [(1-x)^{\alpha+n} (1+x)^{\beta+n}].$$

Recurrence formula:

$$(11) \quad \begin{aligned} & 2(n+1)(n + \alpha + \beta + 1)(2n + \alpha + \beta) P_{n+1}^{(\alpha, \beta)}(x) \\ &= (2n + \alpha + \beta + 1)[(2n + \alpha + \beta)(2n + \alpha + \beta + 2)x + \alpha^2 - \beta^2] P_n^{(\alpha, \beta)}(x) \\ & - 2(n + \alpha)(n + \beta)(2n + \alpha + \beta + 2) P_{n-1}^{(\alpha, \beta)}(x). \end{aligned}$$

From (10) we obtain the explicit expression

$$(12) \quad P_n^{(\alpha, \beta)}(x) = 2^{-n} \sum_{m=0}^n \binom{n+\alpha}{m} \binom{n+\beta}{n-m} (x-1)^{n-m} (x+1)^m$$

which shows that

$$(13) \quad P_n^{(\alpha, \beta)}(-x) = (-1)^n P_n^{(\beta, \alpha)}(x).$$

Differential equation:

$$(14) \quad (1-x^2) y'' + [\beta - \alpha - (\alpha + \beta + 2)x] y' + n(n + \alpha + \beta + 1) y = 0.$$

Differentiation formula:

$$(15) \quad (2n + \alpha + \beta)(1 - x^2) \frac{d}{dx} P_n^{(\alpha, \beta)}(x) \\ = n[(\alpha - \beta) - (2n + \alpha + \beta)x] P_n^{(\alpha, \beta)}(x) + 2(n + \alpha)(n + \beta) P_{n-1}^{(\alpha, \beta)}(x).$$

(iv) *Hypergeometric functions.* Equation (14) can be reduced to the hypergeometric differential equation 2.1(1), and the Jacobi polynomial is that solution of (14) which is regular and has the value (3) at $x = 1$. From the formulas of sec. 2.9,

$$(16) \quad P_n^{(\alpha, \beta)}(x) = \binom{n + \alpha}{n} F(-n, n + \alpha + \beta + 1; \alpha + 1; \tfrac{1}{2} - \tfrac{1}{2}x) \\ = (-1)^n \binom{n + \beta}{n} F(-n, n + \alpha + \beta + 1; \beta + 1; \tfrac{1}{2} + \tfrac{1}{2}x) \\ = \binom{n + \alpha}{n} (\tfrac{1}{2} + \tfrac{1}{2}x)^n F\left(-n, -n - \beta; \alpha + 1; \frac{x - 1}{x + 1}\right) \\ = \binom{n + \beta}{n} (\tfrac{1}{2}x - \tfrac{1}{2})^n F\left(-n, -n - \alpha; \beta + 1; \frac{x + 1}{x - 1}\right).$$

From this we find the further differentiation formula

$$(17) \quad 2^m D^m P_n^{(\alpha, \beta)}(x) = (n + \alpha + \beta + 1)_m P_{n-m}^{(\alpha+m, \beta+m)}(x) \\ m = 1, 2, \dots, n$$

which confirms statement (i) of sec. 10.6.

It follows from 2.9(14) that the function $Q_n^{(\alpha, \beta)}(x)$ defined by

$$(18) \quad \Gamma(2n + \alpha + \beta + 2) Q_n^{(\alpha, \beta)}(x) = \frac{2^{n+\alpha+\beta} \Gamma(n + \alpha + 1) \Gamma(n + \beta + 1)}{(x - 1)^{n+\alpha+1} (x + 1)^\beta} \\ \times F[n + 1, n + \alpha + 1; 2n + \alpha + \beta + 2; 2(1 - x)^{-1}]$$

is a second solution of (14). It is known as the *Jacobi function of the second kind*. This function is not a polynomial, but it satisfies the same recurrence formula (11), and the differentiation formula (15), as the Jacobi polynomial (except that $n = 0$ is not admissible with the Q); it vanishes at infinity when $\text{Re}(\alpha + \beta) > -n - 1$. For the various transformations of the hypergeometric series in (18), and for its analytic continuation, see sec. 2.1.4.

Jacobi polynomials and Jacobi functions of the second kind are connected by several relations. From the connection between various solutions of the hypergeometric equation, see sec. 2.9, we have

$$\begin{aligned}
 (19) \quad Q_n^{(\alpha, \beta)}(x) &= -\frac{1}{2}\pi \operatorname{cosec}(a\pi) P_n^{(\alpha, \beta)}(x) \\
 &\quad + 2^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(n+\beta+1)}{\Gamma(n+\alpha+\beta+1)} (x-1)^{-\alpha} (x+1)^{-\beta} \\
 &\quad \times F(n+1, -n-a-\beta; 1-a; \frac{1}{2}-\frac{1}{2}x).
 \end{aligned}$$

There is also the integral relation

$$(20) \quad Q_n^{(\alpha, \beta)}(x) = \frac{1}{2}(x-1)^{-\alpha} (x+1)^{-\beta} \int_{-1}^1 (x-t)^{-1} (1-t)^{\alpha} (1+t)^{\beta} P_n^{(\alpha, \beta)}(t) dt$$

valid for all points x in the complex plane cut along the segment $(-1, 1)$. This segment is a branchcut, and $Q_n^{(\alpha, \beta)}$ assumes different values according as x approaches a point ξ on the branchcut from the upper half-plane ($\xi+i0$) or from the lower half-plane ($\xi-i0$). The values of $Q_n^{(\alpha, \beta)}(\xi \pm i0)$ may be computed from (19), taking $\arg(x-1) = \pi$ for $x = \xi + i0$, and $\arg(x-1) = -\pi$ for $x = \xi - i0$. In particular,

$$\begin{aligned}
 (21) \quad Q_n^{(\alpha, \beta)}(\xi + i0) - Q_n^{(\alpha, \beta)}(\xi - i0) \\
 &= -i 2^{\alpha+\beta} \sin(a\pi) \frac{\Gamma(\alpha)\Gamma(n+\beta+1)}{\Gamma(n+\alpha+\beta+1)} (1-\xi)^{-\alpha} (1+\xi)^{-\beta} \\
 &\quad \times F(n+1, -n-a-\beta; 1-a; \frac{1}{2}-\frac{1}{2}\xi) \quad -1 < \xi < 1.
 \end{aligned}$$

On the cut itself, one may use the function

$$(22) \quad Q_n^{(\alpha, \beta)}(\xi) = \frac{1}{2}[Q_n^{(\alpha, \beta)}(\xi + i0) + Q_n^{(\alpha, \beta)}(\xi - i0)] \quad -1 < \xi < 1$$

which is real when α and β are real. From (19)

$$\begin{aligned}
 (23) \quad Q_n^{(\alpha, \beta)}(\xi) &= -\frac{1}{2}\pi \operatorname{cosec}(a\pi) P_n^{(\alpha, \beta)}(\xi) \\
 &\quad + 2^{\alpha+\beta-1} \cos(a\pi) \frac{\Gamma(\alpha)\Gamma(n+\beta+1)}{\Gamma(n+\alpha+\beta+1)} (1-\xi)^{-\alpha} (1+\xi)^{-\beta} \\
 &\quad \times F(n+1, -n-a-\beta; 1-a; \frac{1}{2}-\frac{1}{2}\xi) \quad -1 < \xi < 1.
 \end{aligned}$$

Jacobi functions of the second kind are also connected with the polynomials

$$(24) \quad q_n^{(\alpha, \beta)}(x) = \int_{-1}^1 (t-x)^{-1} (1-t)^{\alpha} (1+t)^{\beta} [P_n^{(\alpha, \beta)}(t) - P_n^{(\alpha, \beta)}(x)] dt$$

associated with Jacobi polynomials according to 10.5(6), for clearly (20) may be rewritten as

$$(25) \quad Q_n^{(\alpha, \beta)}(x) = -\frac{1}{2}(x-1)^{-\alpha} (x+1)^{-\beta} q_n^{(\alpha, \beta)}(x) + Q_0^{(\alpha, \beta)}(x) P_n^{(\alpha, \beta)}(x).$$

Other relations connecting the P and Q are

$$(26) \quad P_n^{(\alpha, \beta)}(x) Q_{n-1}^{(\alpha, \beta)}(x) - P_{n-1}^{(\alpha, \beta)}(x) Q_n^{(\alpha, \beta)}(x) \\ = 2^{\alpha+\beta-1} (2n + \alpha + \beta) \frac{\Gamma(\alpha + n) \Gamma(\beta + n)}{n! \Gamma(n + \alpha + \beta + 1)} (x - 1)^{-\alpha} (x + 1)^{-\beta}$$

$$(27) \quad P_n^{(\alpha, \beta)}(x) \frac{d}{dx} Q_n^{(\alpha, \beta)}(x) - Q_n^{(\alpha, \beta)}(x) \frac{d}{dx} P_n^{(\alpha, \beta)}(x) \\ = -2^{\alpha+\beta} \frac{\Gamma(n + \alpha + 1) \Gamma(n + \beta + 1)}{n! \Gamma(n + \alpha + \beta + 1)} (x - 1)^{-\alpha-1} (x + 1)^{-\beta-1},$$

and from these it follows that $Q_n^{(\alpha, \beta)}$ satisfies the same differentiation formula (15) as $P_n^{(\alpha, \beta)}$.

From the theory of hypergeometric functions one obtains *integral representations* for $Q_n^{(\alpha, \beta)}$. The simplest of these is

$$(28) \quad Q_n^{(\alpha, \beta)}(x) = 2^{-n-1} (x - 1)^{-\alpha} (x + 1)^{-\beta} \\ \times \int_{-1}^1 (x - t)^{-n-1} (1 - t)^{n+\alpha} (1 + t)^{n+\beta} dt,$$

valid when x is in the complex plane cut along the segment $(-1, 1)$.

(v) *Generating function.*

$$(29) \quad \sum_{n=0}^{\infty} P_n^{(\alpha, \beta)}(x) z^n = 2^{\alpha+\beta} R^{-1} (1 - z + R)^{-\alpha} (1 + z + R)^{-\beta} \quad |z| < 1$$

where

$$(30) \quad R = (1 - 2xz + z^2)^{1/2}$$

and $R = 1$ when $z = 0$. For several ways of proving (29) see Szegő (1939, sec. 4.4). For particular values of α, β there are other generating functions.

(vi) *Integral representations.* From Rodrigues' formula (10), we have

$$(31) \quad P_n^{(\alpha, \beta)}(x) = \frac{1}{2\pi i} \int_{(x+)} \left(\frac{1}{2} \frac{t^2 - 1}{t - x} \right)^n \left(\frac{1 - t}{1 - x} \right)^{\alpha} \left(\frac{1 + t}{1 + x} \right)^{\beta} dt$$

where $x \neq \pm 1$, the contour of integration is a simple closed contour, in the positive sense, around $t = x$. The points $t = \pm 1$ are outside the contour, and $[(1 - t)/(1 - x)]^{\alpha}$ and $[(1 + t)/(1 + x)]^{\beta}$ are to be taken as unity when $t = x$.

Further integral representations may be obtained, from integrals representing hypergeometric functions, by means of (16).

(vii) *Miscellaneous results.* We may apply Christoffel's formula 10.3 (12) to $w(x) = (1-x)^\alpha (1+x)^\beta$, $\rho(x) = (1-x)$. In virtue of (3) we obtain

$$(32) \quad (n + \frac{1}{2}\alpha + \frac{1}{2}\beta + 1)(1-x) P_n^{(\alpha+1, \beta)}(x) \\ = (n + \alpha + 1) P_n^{(\alpha, \beta)}(x) - (n + 1) P_{n+1}^{(\alpha, \beta)}(x)$$

and similarly

$$(33) \quad (n + \frac{1}{2}\alpha + \frac{1}{2}\beta + 1)(1+x) P_n^{(\alpha, \beta+1)}(x) \\ = (n + \beta + 1) P_n^{(\alpha, \beta)}(x) + (n+1) P_{n+1}^{(\alpha, \beta)}(x).$$

These are examples of relations between contiguous hypergeometric functions (see 2.8(31) to 2.8(45)); other relations of this nature are

$$(34) \quad (1-x) P_n^{(\alpha+1, \beta)}(x) + (1+x) P_n^{(\alpha, \beta+1)}(x) = 2 P_n^{(\alpha, \beta)}(x)$$

$$(35) \quad (2n + \alpha + \beta) P_n^{(\alpha-1, \beta)}(x) = (n + \alpha + \beta) P_n^{(\alpha, \beta)}(x) - (n + \beta) P_{n-1}^{(\alpha, \beta)}(x)$$

$$(36) \quad (2n + \alpha + \beta) P_n^{(\alpha, \beta-1)}(x) = (n + \alpha + \beta) P_n^{(\alpha, \beta)}(x) + (n + \alpha) P_{n-1}^{(\alpha, \beta)}(x)$$

$$(37) \quad P_n^{(\alpha, \beta-1)}(x) - P_n^{(\alpha-1, \beta)}(x) = P_{n-1}^{(\alpha, \beta)}(x).$$

Repeated application of these formulas results in the expression of $P_n^{(\alpha+h, \beta+k)}(x)$ for any integers h, k in terms of $P_n^{(\alpha, \beta)}(x)$.

From Rodrigues' formula (10) we have

$$(38) \quad 2n \int_0^x (1-y)^\alpha (1+y)^\beta P_n^{(\alpha, \beta)}(y) dy \\ = P_{n-1}^{(\alpha+1, \beta+1)}(0) - (1-x)^{\alpha+1} (1+x)^{\beta+1} P_{n-1}^{(\alpha+1, \beta+1)}(x).$$

Toscano (1949) found a counterpart of Rodrigues' formula in terms of finite differences. We define the difference operator by

$$(39) \quad \Delta_\alpha F(\alpha) = F(\alpha+1) - F(\alpha), \quad \Delta_\alpha^n F = \Delta_\alpha(\Delta_\alpha^{n-1} F)$$

and write Toscano's result in the form

$$(40) \quad n! \Gamma(\alpha + \beta + n + 1) P_n^{(\alpha, \beta)}(x) \\ = \frac{(-1)^n \Gamma(\alpha + n + 1)}{(\frac{1}{2} - \frac{1}{2}x)^{\alpha+1}} \Delta_\alpha^n \left[\frac{\Gamma(\alpha + \beta + n + 1)}{\Gamma(\alpha + 1)} (\frac{1}{2} - \frac{1}{2}x)^{\alpha+1} \right].$$

Lastly we quote the important limit

$$(41) \quad \lim_{n \rightarrow \infty} \left[n^{-\alpha} P_n^{(\alpha, \beta)} \left(\cos \frac{z}{n} \right) \right] = \lim_{n \rightarrow \infty} \left[n^{-\alpha} P_n^{(\alpha, \beta)} \left(1 - \frac{z^2}{2n^2} \right) \right] \\ = (\frac{1}{2}z)^{-\alpha} J_\alpha(z)$$

where J_a is the Bessel function of the first kind. This formula holds for arbitrary a and β , uniformly in any bounded region of the complex z -plane.

10.9. Gegenbauer polynomials

We use Gegenbauer's notation $C_n^\lambda(x)$ for the suitably standardized polynomials associated with

$$(1) \quad a = -1, \quad b = 1, \quad w(x) = (1 - x^2)^{\lambda - 1/2}, \quad X = 1 - x^2.$$

These polynomials are also known as *ultraspherical* polynomials and are often denoted by $P_n^{(\lambda)}(x)$. Clearly, Gegenbauer polynomials are constant multiples of Jacobi polynomials with $\alpha = \beta = \lambda - 1/2$. In order to have a real and integrable weight function we assume

$$(2) \quad \lambda > -1/2,$$

although many of the formal relations are valid without this restriction. For these polynomials see also sec. 3.15.

(i) *Standardization.*

$$(3) \quad C_n^\lambda(1) = \binom{n + 2\lambda - 1}{n} = \frac{(2\lambda)_n}{n!}.$$

By comparison with 10.8(3)

$$(4) \quad (\lambda + 1/2)_n C_n^\lambda(x) = (2\lambda)_n P_n^{(\alpha, \alpha)}(x) \quad \alpha = \lambda - 1/2.$$

The standardization (3) fails when 2λ is zero or a negative integer. The only exception in the range (2) is $\lambda = 0$ and for this we standardize according to

$$(5) \quad C_0^0(1) = 1, \quad C_n^0(1) = \frac{2}{n} \quad n = 1, 2, \dots,$$

and have

$$(6) \quad C_n^0(x) = \lim_{\lambda \rightarrow \infty} \lambda^{-1} C_n^\lambda(x) = 2 \frac{(n-1)!}{(1/2)_n} P_n^{(-1/2, -1/2)}(x).$$

In many formulas of this section $\lambda = 0$ must be excluded. This case will be considered in sec. 10.10.

(ii) *Constants.*

$$(7) \quad (n + \lambda) n! \Gamma(\lambda) h_n = \pi^{1/2} (2\lambda)_n \Gamma(\lambda + 1/2)$$

$$(8) \quad n! k_n = 2^n (\lambda)_n, \quad r_n = 0, \quad (2\lambda)_n K_n = (-2)^n (\lambda + 1/2)_n$$

$$(9) \quad (n + 1) A_n = 2(n + \lambda), \quad B_n = 0, \quad (n + 1) C_n = n + 2\lambda - 1$$

$$(10) \quad \lambda_n = n(n + 2\lambda), \quad \alpha_n = 0, \quad \beta_n = n + 2\lambda - 1.$$

(iii) *Rodrigues' formula.*

$$(11) \quad 2^n n! (\lambda + \frac{1}{2})_n (1 - x^2)^{\lambda - \frac{1}{2}} C_n^\lambda(x) = (-1)^n (2\lambda)_n D^n [(1 - x^2)^{n + \lambda - \frac{1}{2}}]$$

$$(12) \quad C_0^\lambda(x) = 1, \quad C_1^\lambda(x) = 2\lambda x.$$

Recurrence formula

$$(13) \quad (n + 1) C_{n+1}^\lambda(x) = 2(n + \lambda) x C_n^\lambda(x) - (n + 2\lambda - 1) C_{n-1}^\lambda(x).$$

Differential equation

$$(14) \quad (1 - x^2) y'' - (2\lambda + 1) xy' + n(n + 2\lambda) y = 0.$$

Differentiation formula

$$(15) \quad (1 - x^2) \frac{d}{dx} C_n^\lambda(x) = -nx C_n^\lambda(x) + (n + 2\lambda - 1) C_{n-1}^\lambda(x) \\ = (n + 2\lambda) x C_n^\lambda(x) - (n + 1) C_{n+1}^\lambda(x).$$

Parity

$$(16) \quad C_n^\lambda(-x) = (-1)^n C_n^\lambda(x).$$

Explicit representations

$$(17) \quad C_n^\lambda(\cos \theta) = \sum_{m=0}^n \frac{(\lambda)_m (\lambda)_{n-m}}{m! (n-m)!} \cos(n-2m)\theta$$

$$(18) \quad C_n^\lambda(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m (\lambda)_{n-m}}{m! (n-2m)!} (2x)^{n-2m}$$

$$(19) \quad C_n^\lambda(0) = \begin{cases} 0 & \text{if } n \text{ is odd,} \\ (-1)^m (\lambda)_m / m! & \text{if } n = 2m \text{ is even.} \end{cases}$$

(iv) *Hypergeometric functions.* The differential equation (14) can be reduced to the hypergeometric equation, and $C_n^\lambda(x)$ is that solution which is regular at $x = 1$ and has the value (3) there. Moreover, in the case of Gegenbauer polynomials the hypergeometric series in question admit of quadratic transformation, see sec. 2.1.5, and we obtain the following representations:

$$\begin{aligned}
 (20) \quad n! C_n^\lambda(x) &= (2\lambda)_n F(-n, n+2\lambda; \lambda+\frac{1}{2}; \frac{1}{2}-\frac{1}{2}x) \\
 &= (-1)^n (2\lambda)_n F(-n, n+2\lambda; \lambda+\frac{1}{2}; \frac{1}{2}+\frac{1}{2}x) \\
 &= 2^n (\lambda)_n (x-1)^n F\left(-n, -n-\lambda+\frac{1}{2}; -2n-2\lambda+1; \frac{2}{1-x}\right) \\
 &= (2\lambda)_n \left(\frac{1}{2}+\frac{1}{2}x\right)^n F\left(-n, -n-\lambda+\frac{1}{2}; \lambda+\frac{1}{2}; \frac{x-1}{x+1}\right)
 \end{aligned}$$

$$\begin{aligned}
 (21) \quad C_{2m}^\lambda(x) &= (-1)^m \frac{(\lambda)_m}{m!} F(-m, m+\lambda; \frac{1}{2}; x^2) \\
 &= \frac{(2\lambda)_{2m}}{(2m)!} F(-m, m+\lambda; \lambda+\frac{1}{2}; 1-x^2) \\
 &= \frac{(\lambda)_m}{(\frac{1}{2})_m} P_m^{\lambda-\frac{1}{2}, -\frac{1}{2}}(2x^2-1)
 \end{aligned}$$

$$\begin{aligned}
 (22) \quad C_{2m+1}^\lambda(x) &= (-1)^m \frac{(\lambda)_{m+1}}{m!} 2x F\left(-m, m+\lambda+1; \frac{3}{2}; x^2\right) \\
 &= \frac{(2\lambda)_{2m+1}}{(2m+1)!} x F(-m, m+\lambda+1; \lambda+\frac{1}{2}; 1-x^2) \\
 &= \frac{(\lambda)_{m+1}}{(\frac{1}{2})_{m+1}} x P_m^{\lambda-\frac{1}{2}, \frac{1}{2}}(2x^2-1).
 \end{aligned}$$

From these representations in conjunction with (13) and (19) one obtains

$$(23) \quad D^m C_n^\lambda(x) = 2^m (\lambda)_m C_{n-m}^{\lambda+m}(x) \quad m = 1, 2, \dots, n$$

$$(24) \quad D C_{n-1}^\lambda(x) = x D C_n^\lambda(x) - n C_n^\lambda(x)$$

$$(25) \quad D C_{n+1}^\lambda(x) = x D C_n^\lambda(x) + (n+2\lambda) C_n^\lambda(x)$$

$$(26) \quad 2(n+\lambda) \int C_n^\lambda(x) dx = C_{n+1}^\lambda(x) - C_{n-1}^\lambda(x)$$

$$(27) \quad D C_n^\lambda(0) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ 2(-1)^m (\lambda)_{m+1}/m! & \text{if } n = 2m+1 \text{ is odd.} \end{cases}$$

A second solution of the differential equation (14) can be obtained from the work of 10.8(iv) by means of the connection, (4), (6), (21), or (22), between Gegenbauer and Jacobi polynomials. No generally accepted notation or standardization seems to exist in this case.

(v) *Generating functions.* From 10.8(29),

$$(28) \quad \sum_{n=0}^{\infty} \frac{(\lambda + \frac{1}{2})_n}{(2\lambda)_n} C_n^\lambda(x) z^n = 2^{\lambda-\frac{1}{2}} R^{-1} (1 - xz + R)^{\frac{1}{2}-\lambda}$$

$$|z| < 1, \quad R = (1 - 2xz + z^2)^{\frac{1}{2}}, \quad R = 1 \text{ when } z = 0;$$

but in this case there is a simpler generating function, viz.

$$(29) \quad \sum_{n=0}^{\infty} C_n^\lambda(x) z^n = (1 - 2xz + z^2)^{-\lambda} \quad |z| < 1$$

which can be verified by putting $x = \cos \theta$, factorizing the right-hand side as $(1 - e^{i\theta} z)^{-\lambda} (1 - e^{-i\theta} z)^{-\lambda}$, expanding in the binomial series, and using (17). A third generating function

$$(30) \quad \sum_{n=0}^{\infty} C_n^\lambda(x) \frac{z^n}{(2\lambda)_n} = \Gamma(\lambda + \frac{1}{2}) e^{z \cos \theta} (\frac{1}{2} z \sin \theta)^{\frac{1}{2}-\lambda} J_{\lambda-\frac{1}{2}}(z \sin \theta)$$

is connected with (29) by means of the Laplace transformation.

(vi) *Integral representations.* Each of the generating functions leads to a contour integral representation of Gegenbauer polynomials. In addition, we have the real integrals

$$(31) \quad C_n^\lambda(x) = \frac{2^{1-2\lambda} \Gamma(2\lambda + n)}{n! [\Gamma(\lambda)]^2} \int_0^\pi [x + (x^2 - 1)^{\frac{1}{2}} \cos \varphi]^n (\sin \varphi)^{2\lambda-1} d\varphi$$

$$(32) \quad C_n^\lambda(\cos \theta) = \frac{2^\lambda \Gamma(\lambda + \frac{1}{2}) (2\lambda)_n}{\pi^{\frac{1}{2}} n! \Gamma(\lambda)} (\sin \theta)^{1-2\lambda} \int_0^\theta \frac{\cos(n + \lambda)\varphi}{(\cos \varphi - \cos \theta)^{1-\lambda}} d\varphi,$$

both for $\lambda > 0$. For (31) see 3.15(22) and Seidel and Szász (1950). Equation (32) is *Mehler's integral* 3.15(23); there is a second integral obtained by replacing ϕ and θ by $\pi - \phi$ and $\pi - \theta$ respectively. Mehler's integral suggests a functional transformation which will carry ultraspherical polynomials into powers.

(vii) *Miscellaneous results.* From the connection with Legendre functions,

$$(33) \quad n! C_n^\lambda(x) = \Gamma(\lambda + \frac{1}{2}) (2\lambda)_n [\frac{1}{4}(x^2 - 1)]^{\frac{1}{2}-\lambda} P_{n+\lambda-\frac{1}{2}}^{\frac{1}{2}-\lambda}(x),$$

we have the addition theorem

$$\begin{aligned}
 (34) \quad & C_n^\lambda(\cos \theta \cos \psi + \sin \theta \sin \psi \cos \varphi) \\
 &= \sum_{m=0}^n 2^m (2\lambda + 2m - 1)(n - m)! \frac{[(\lambda)_m]^2}{(2\lambda - 1)_{n+m+1}} \\
 &\quad \times (\sin \theta)^m C_{n-m}^{\lambda+m}(\cos \theta) (\sin \psi)^m C_{n-m}^{\lambda+m}(\cos \psi) C_m^{\lambda-\frac{1}{2}}(\cos \varphi).
 \end{aligned}$$

Relations between contiguous hypergeometric functions are

$$\begin{aligned}
 (35) \quad & 2\lambda(1-x^2) C_{n-1}^{\lambda+1}(x) = (2\lambda + n - 1) C_{n-1}^\lambda(x) - nx C_n^\lambda(x) \\
 &= (n + 2\lambda)x C_n^\lambda(x) - (n + 1) C_{n+1}^\lambda(x)
 \end{aligned}$$

$$(36) \quad (n + \lambda) C_{n+1}^{\lambda-1}(x) = (\lambda - 1) [C_{n+1}^\lambda(x) - C_{n-1}^\lambda(x)].$$

The differentiation formula

$$(37) \quad (x^2 - 1)^{\lambda + \frac{1}{2}n} D^n [(x^2 - 1)^{-\lambda}] = (-1)^n n! C_n^\lambda [x(x^2 - 1)^{-\frac{1}{2}}]$$

follows from (11) and a linear transformation of the hypergeometric series in (21) and (22). It is due to Tricomi (1949). We note also Gegenbauer's integral

$$\begin{aligned}
 (38) \quad & n! \int_0^\pi e^{iz \cos \theta} C_n^\lambda(\cos \theta) (\sin \theta)^{2\lambda} d\theta \\
 &= 2^\lambda \pi^{\frac{1}{2}} \Gamma(\lambda + \frac{1}{2})(2\lambda)_n i^n z^{-\lambda} J_{\lambda+n}(z)
 \end{aligned}$$

and the expansion in a trigonometric series

$$\begin{aligned}
 (39) \quad & \Gamma(\lambda) C_n^\lambda(\cos \theta) = 2 \sum_{m=0}^{\infty} \frac{(\lambda)_m}{m!} \frac{\Gamma(n+m+2\lambda)}{\Gamma(n+m+\lambda+1)} \cos[(n+2m+2\lambda)\theta - \lambda\pi] \\
 & \qquad \qquad \qquad 0 < \lambda < 1, \quad 0 < \theta < \pi
 \end{aligned}$$

(Szegő, 1939, p. 95).

10.10. Legendre polynomials

Legendre polynomials $P_n(x)$ are the suitably standardized polynomials associated with

$$(1) \quad a = -1, \quad b = 1, \quad w(x) = 1, \quad X = 1 - x^2.$$

These polynomials are also known as *spherical polynomials*. Clearly they are Jacobi polynomials with $\alpha = \beta = 0$, and also Gegenbauer polynomials with $\lambda = \frac{1}{2}$. Legendre polynomials, and more generally Legendre functions have been studied extensively (cf. chapter III).

(i) *Standardization.*

$$(2) \quad P_n(1) = 1.$$

Hence

$$(3) \quad P_n(x) = C_n^{1/2}(x) = P_n^{(0,0)}(x).$$

(ii) *Constants.*

$$(4) \quad h_n = (n + 1/2)^{-1}, \quad k_n = 2^n g_n = 2^n \frac{(1/2)_n}{n!}, \quad r_n = 0$$

$$(5) \quad K_n = (-2)^n n!, \quad (n+1)A_n = 2n+1, \quad B_n = 0, \quad (n+1)C_n = -n$$

$$(6) \quad \lambda_n = n(n+1), \quad \alpha_n = 0, \quad \beta_n = n.$$

(iii) *Rodrigues' formula.*

$$(7) \quad 2^n n! P_n(x) = D^n [(x^2 - 1)^n]$$

$$(8) \quad P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}.$$

Recurrence formula

$$(9) \quad (n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Christoffel-Darboux formula

$$(10) \quad \sum_{m=0}^n (2m+1) P_m(x) P_m(y) = \frac{n+1}{x-y} [P_{n+1}(x) P_n(y) - P_n(x) P_{n+1}(y)].$$

Differential equation

$$(11) \quad (1-x^2)y'' - 2xy' + n(n+1)y = 0.$$

Differentiation and integration formulas

$$(12) \quad (1-x^2)P'_n(x) = n[P_{n-1}(x) - xP_n(x)] = (n+1)[xP_n(x) - P_{n+1}(x)]$$

$$(13) \quad xP'_n(x) - P'_{n-1}(x) = nP_n(x)$$

$$(14) \quad P'_{n+1}(x) - xP'_n(x) = (n+1)P_n(x)$$

$$(15) \quad (2n+1) \int P_n(x) dx = P_{n+1}(x) - P_{n-1}(x).$$

In these formulas $P'_n(x) = dP_n(x)/dx$.

Explicit representations, parity, special values

$$(16) \quad P_n(x) = 2^{-n} \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m \binom{n}{m} \binom{2n-2m}{n} x^{n-2m}$$

$$(17) \quad P_n(\cos \theta) = \sum_{m=0}^n g_m g_{n-m} \cos(n-2m)\theta$$

$$(18) \quad P_n(-x) = (-1)^n P_n(x), \quad P_n(\pm 1) = (\pm 1)^n$$

$$(19) \quad P_{2m}(0) = (-1)^m g_m, \quad P_{2m+1}(0) = 0$$

$$(20) \quad P'_{2m}(0) = 0, \quad P'_{2m+1}(0) = (-1)^m (2m+1) g_m.$$

Here

$$(21) \quad g_m = \frac{(\frac{1}{2})_m}{m!} = 2^{-2m} \binom{2m}{m}.$$

(iv) *Hypergeometric functions.* See also 10.9(iv).

$$(22) \quad P_n(x) = F(-n, n+1; 1; \frac{1}{2} - \frac{1}{2}x) \\ = 2^n g_n x^n F(-\frac{1}{2}n, \frac{1}{2} - \frac{1}{2}n; \frac{1}{2}; x^{-2})$$

$$(23) \quad P_n(\cos \theta) = F(-n, n+1; 1; \sin^2 \frac{1}{2}\theta) = (-1)^n F(-n, n+1; 1; \cos^2 \frac{1}{2}\theta)$$

$$(24) \quad P_{2m}(x) = (-1)^m g_m F(-m, m + \frac{1}{2}; \frac{1}{2}; x^2)$$

$$(25) \quad P_{2m+1}(x) = (-1)^m (2m+1) g_m x F\left(-m, m + \frac{3}{2}; \frac{3}{2}; x^2\right)$$

$$(26) \quad \frac{d^m}{dx^m} P_n(x) = 2^m m! g_m C_{n-m}^{m+\frac{1}{2}}(x) \quad n \geq m.$$

Information about a second solution of Legendre's differential equation (11) may be obtained from 10.8(iv). Such a second solution is the *Legendre function of the second kind*

$$(27) \quad Q_n(x) = Q_n^{(0,0)}(x).$$

In the complex x -plane cut along the segment $(-1, 1)$

$$(28) \quad 2^{-n} (2n+1)! (n!)^{-2} Q_n(x) \\ = (x-1)^{-n-1} F[n+1, n+1; 2n+2; 2(1-x)^{-1}] \\ = (x+1)^{-n-1} F[n+1, n+1; 2n+2; 2(1+x)^{-1}] \\ = x^{-n-1} F(1/2+n/2, 1+n/2; 3/2+n; x^{-2}).$$

The Legendre function of the second kind is not a polynomial: it satisfies the same recurrence relation (9) and the same differentiation formulas (12) - (15) as the Legendre polynomials, except that $n = 0$ is inadmissible in these formulas when written for Q .

$$(29) \quad Q_n(-x) = (-1)^{n+1} Q_n(x)$$

$$(30) \quad Q_0(x) = \frac{1}{2} \log \frac{x+1}{x-1}, \quad Q_1(x) = \frac{1}{2} x \log \frac{x+1}{x-1} - 1$$

$$(31) \quad Q_n(x) = 2^{-n-1} \int_{-1}^1 (1-t^2)^n (x-t)^{-n-1} dt$$

$$(32) \quad Q_n(x) = \int_0^\infty [x + (x^2 - 1)^{1/2} \cosh t]^{-n-1} dt$$

$$(33) \quad Q_n(\cosh \zeta) = \int_\zeta^\infty [2(\cosh z - \cosh \zeta)]^{-1/2} e^{-(n+1/2)z} dz$$

$$\operatorname{Re} z \geq \operatorname{Re} \zeta, \quad \operatorname{Im} z = \operatorname{Im} \zeta$$

$$(34) \quad Q_n(x) = \frac{1}{2} \int_{-1}^1 (x-t)^{-1} P_n(t) dt.$$

$$(35) \quad Q_n(x) = Q_0(x) P_n(x) - \sum_{k=1}^{[(n+1)/2]} \frac{2n-4k+3}{(2k-1)(n-k+1)} P_{n-2k+1}(x).$$

The last formula is equivalent to the special case $\alpha = \beta = 0$ of 10.8.25; for the proof in the form (35) see Hobson (1931, pp. 53-54). The point at infinity is a zero of multiplicity $n+1$ of $Q_n(x)$; and this function has no other zero in the cut x -plane.

The segment of the real axis from -1 to 1 is a branchcut of $Q_n(x)$, and

$$(36) \quad Q_n(\xi + i0) - Q_n(\xi - i0) = -\pi i P_n(\xi) \quad -1 < \xi < 1.$$

On the branchcut we may define a second solution of Legendre's equation by

$$(37) \quad Q_n(\xi) = \frac{1}{2} Q_n(\xi + i0) + \frac{1}{2} Q_n(\xi - i0) \quad -1 < \xi < 1.$$

We then have

$$(38) \quad Q_n(\xi) = \frac{1}{2} \int_{-1}^1 (\xi - t)^{-1} P_n(t) dt \quad -1 < \xi < 1$$

where the integral is a Cauchy principal value, that is

$$\lim \left(\int_{-1}^{\xi-\epsilon} + \int_{\xi+\epsilon}^1 \right) \quad \text{as} \quad \epsilon > 0, \quad \epsilon \rightarrow 0.$$

(v) *Generating functions.*

$$(39) \quad \sum_{n=0}^{\infty} P_n(x) z^n = (1 - 2xz + z^2)^{-1/2} \quad -1 < x < 1, \quad |z| < 1$$

$$(40) \quad \sum_{n=0}^{\infty} \frac{1}{n!} P_n(\cos \theta) z^n = e^{z \cos \theta} J_0(z \sin \theta)$$

$$(41) \quad \sum_{n=0}^{\infty} \frac{(-1)^n}{n + \frac{1}{2}} P_n(\cos \theta) x^{2n+1} = F(\sin \frac{1}{2} \theta, \varphi)$$

$$x = \tan \frac{1}{2} \varphi \quad 0 < \varphi < \frac{1}{2} \pi, \quad 0 < \theta < \pi.$$

The first two formulas are particular cases of 10.9(29) and 10.9(30). The last formula may be derived from (39), and $F(k, \varphi)$ denotes Legendre's incomplete elliptic integral of the first kind with modulus k .

(vi) *Integral representations.*

$$(42) \quad P_n(\cos \theta) = \pi^{-1} \int_0^\pi (\cos \theta + i \sin \theta \cos \varphi)^n d\varphi$$

$$= \pi^{-1} \int_0^\pi (\cos \theta + i \sin \theta \cos \varphi)^{-n-1} d\varphi$$

$$(43) \quad P_n(\cos \theta) = 2^{1/2} \pi^{-1} \int_0^\theta (\cos \varphi - \cos \theta)^{-1/2} \cos(n + \frac{1}{2}) \varphi d\varphi \quad 0 < \theta < \pi$$

$$(44) \quad P_n(x) = (2\pi i)^{-1} \int^{(0+)} (1 - 2xz + z^2)^{-1/2} z^{-n-1} dz$$

$$(45) \quad P_n(x) = (-2)^{-n} (2\pi i)^{-1} \int^{(x+)} (1 - z^2)^n (z - x)^{-n-1} dz.$$

Equation (44) follows from (39), and (45) from Rodrigues' formula. The integral in (45) is known as Schläfli's integral. Laplace's first and second integral, (42), may be deduced from (45) when the contour of integration is taken to be the circle

$$z = x + (x^2 - 1)^{1/2} e^{i\varphi} \quad -\pi \leq \varphi < \pi,$$

and Mehler's integral (43) may be deduced from Laplace's integral (Whittaker and Watson 1940, sections 15.23 and 15.231).

(vii) *Miscellaneous results.* With the notation

$$(46) \quad P_n^m(\cos \theta) = (-2)^m m! g_m(\sin \theta)^m C_{n-m}^{m+1/2}(\cos \theta)$$

for the *associated Legendre function of the first kind* [see 3.4(1) and 3.15(4)] we have from 10.9(34) the *addition theorem* of Legendre polynomials

$$(47) \quad P_n(\cos \theta \cos \psi + \sin \theta \sin \psi \cos \varphi) = P_n(\cos \theta) P_n(\cos \psi) \\ + 2 \sum_{m=1}^n \frac{(n-m)!}{(n+m)!} P_m^*(\cos \theta) P_m^*(\cos \psi) \cos n \varphi.$$

We note the expansion in a trigonometric series

$$(48) \quad P_{n-1}(\cos \theta) = \frac{2}{\pi n g_n} \sum_{m=0}^{\infty} \frac{(n)_m g_m}{(n + \frac{1}{2})_m} \sin[(n+2m)\theta] \quad n = 2, 3, \dots$$

and the integral formulas

$$(49) \quad \int_{-1}^1 (1-x)^{-1/2} P_n(x) dx = \frac{2^{3/2}}{2n+1}$$

$$(50) \quad \int_0^\pi P_{2m}(\cos \theta) d\theta = \pi g_m^2, \quad \int_0^\pi P_{2m+1}(\cos \theta) \cos \theta d\theta = \pi g_m g_{m+1}$$

$$(51) \quad \int_0^1 x^\lambda P_{2m}(x) dx = \frac{(-1)^m (-\frac{1}{2}\lambda)_m}{2(\frac{1}{2} + \frac{1}{2}\lambda)_{m+1}} \quad \operatorname{Re} \lambda > -1$$

$$(52) \quad \int_0^1 x^\lambda P_{2m+1}(x) dx = \frac{(-1)^m (\frac{1}{2} - \frac{1}{2}\lambda)_m}{2(1 + \frac{1}{2}\lambda)_{m+1}} \quad \operatorname{Re} \lambda > -2$$

and the bilinear expansion

$$(53) \quad \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} P_n(x) P_n(y) = 2 \log 2 - 1 - \log[(1-x)(1+y)] \\ -1 < x \leq y < 1.$$

10.11. Tchebichef polynomials

Sometimes (especially in the French literature) orthogonal polynomials in general are called Tchebichef polynomials. There are also several special systems of orthogonal polynomials called Tchebichef polynomials. In this chapter we shall reserve the name Tchebichef polynomials of the first and second kind for the suitably standardized orthogonal polynomials associated with

$$(1) \quad a = -1, \quad b = 1, \quad w(x) = (1-x^2)^{-1/2}, \quad X = 1-x^2.$$

Clearly these polynomials are multiples of Jacobi polynomials with $a = \beta = -\frac{1}{2}$ for the polynomials of the first kind $T_n(x)$ and with $a = \beta = \frac{1}{2}$ for the polynomials of the second kind $U_n(x)$. Also, the Jacobi polynomials in question are ultraspherical polynomials with $\lambda = 0$ for the

polynomials of the first kind and with $\lambda = 1$ for the polynomials of the second kind.

The orthogonal relationship for Tchebichef polynomials of the first kind reads

$$\int_{-1}^1 T_m(x) T_n(x) (1-x^2)^{-1/2} dx = 0 \quad m \neq n.$$

If we substitute $x = \cos \theta$ and note that $\cos n \theta$ is a polynomial of exact degree n in $\cos \theta$, we see that $T_n(x)$ must be a constant multiple of $\cos n \theta$; and we show in a similar manner that $U_n(x)$ is a constant multiple of $\csc \theta \sin(n+1)\theta$. We standardize our polynomials by putting

$$(2) \quad T_n(\cos \theta) = \cos n \theta, \quad U_n(\cos \theta) = \frac{\sin(n+1)\theta}{\sin \theta}.$$

Many identities involving Tchebichef polynomials are paraphrases of well-known trigonometric identities. As an example, we mention the connection between the two kinds of Tchebichef polynomials,

$$(3) \quad T_n(x) = U_n(x) - x U_{n-1}(x)$$

$$(4) \quad (1-x^2) U_{n-1}(x) = x T_n(x) - T_{n+1}(x).$$

Tchebichef polynomials are ultraspherical polynomials with $\lambda = 0, 1$. From 10.9(23) it is seen that $C_n^\lambda(x)$ can be expressed as a derivative of a Tchebichef polynomial whenever λ is a positive integer.

(i) *Standardization.* This is given by (2). It follows that

$$(5) \quad T_n(x) = \frac{1}{2} n C_n^0(x) = (g_n)^{-1} P_n^{(-1/2, -1/2)}(x) \quad n = 1, 2, \dots$$

$$(6) \quad U_n(x) = C_n^1(x) = (2g_{n+1})^{-1} P_n^{(1/2, 1/2)}(x) \quad n = 0, 1, \dots$$

where C_n^0 is defined by 10.9(6) and g_n by 10.10(21).

(ii) *Constants.* For $T_n(x)$

$$(7) \quad h_0 = \pi, \quad h_n = \frac{1}{2} \pi \quad n = 1, 2, \dots$$

$$(8) \quad k_n = 2^{n-1}, \quad r_n = 0, \quad K_n = (-1)^n 2^n n! g_n$$

$$(9) \quad A_n = 2, \quad B_n = 0, \quad C_n = 1$$

$$(10) \quad \lambda_n = n^2, \quad \alpha_n = 0, \quad \beta_n = n.$$

For $U_n(x)$

$$(11) \quad h_n = \frac{1}{2}n, \quad k_n = 2^n, \quad r_n = 0, \quad K_n = (-1)^n 2^{n+1} n! g_{n+1}$$

$$(12) \quad A_n = 2, \quad B_n = 0, \quad C_n = 1$$

$$(13) \quad \lambda_n = n(n+2), \quad a_n = 0, \quad \beta_n = n+1.$$

(iii) *Rodrigues' formulas.*

$$(14) \quad 2^n \left(\frac{1}{2}\right)_n T_n(x) = (-1)^n (1-x^2)^{\frac{1}{2}} D^n [(1-x^2)^{n-\frac{1}{2}}]$$

$$(15) \quad 2^{n+1} \left(\frac{1}{2}\right)_{n+1} U_n(x) = (-1)^n (n+1) (1-x^2)^{-\frac{1}{2}} D^n [(1-x^2)^{n+\frac{1}{2}}].$$

Recurrence formula [$z_n(x)$ is either $T_n(x)$ or $U_n(x)$]

$$(16) \quad z_{n+1}(x) = 2xz_n(x) - z_{n-1}(x).$$

Christoffel-Darboux formula.

$$(17) \quad \sum_{m=0}^n {}' z_m(x) z_m(y) = (x-y)^{-1} [z_{n+1}(x) z_n(y) - z_n(x) z_{n+1}(y)]$$

where z_n is either T_n or U_n , but in the case of T_n , the first term ($m=0$) of the sum must be halved.

Differential equations.

$$(18) \quad (1-x)y'' - xy' + n^2y = 0 \quad \text{for} \quad y = T_n(x)$$

$$(19) \quad (1-x)y'' - 3xy' + n(n+2)y = 0 \quad \text{for} \quad y = U_n(x)$$

Differentiation formulas (primes denote differentiation with respect to x)

$$(20) \quad (1-x^2) T_n'(x) = n [T_{n-1}(x) - x T_n(x)]$$

$$(21) \quad (1-x^2) U_n'(x) = (n+1) U_{n-1}(x) - nx U_n(x).$$

Explicit representations

$$(22) \quad T_n(x) = \frac{n}{2} \sum_{m=0}^{[n/2]} \frac{(-1)^m (n-m-1)!}{m! (n-2m)!} (2x)^{n-2m} \quad n = 1, 2, \dots$$

$$(23) \quad U_n(x) = \sum_{m=0}^{[n/2]} \frac{(-1)^m (n-m)!}{m! (n-2m)!} (2x)^{n-2m}.$$

(iv) *Hypergeometric functions.*

$$(24) \quad T_n(x) = F(-n, n; \frac{1}{2}; \frac{1}{2} - \frac{1}{2}x)$$

$$(25) \quad U_n(x) = (n+1) F\left(-n, n+1; \frac{3}{2}; \frac{1}{2} - \frac{x}{2}\right).$$

From these relations and from 10.9(iv)

$$(26) \quad D^m T_n(x) = 2^{m-1} (m-1)! n C_{n-m}^m(x) \quad n \geq m$$

$$(27) \quad D^m U_n(x) = 2^m m! C_{n-m}^{m+1}(x) \quad n \geq m$$

$$(28) \quad T_n'(x) = n U_{n-1}(x).$$

(v) *Generating functions.*

$$(29) \quad 1 + 2 \sum_{n=1}^{\infty} T_n(x) z^n = \frac{1 - z^2}{1 - 2xz + z^2}$$

$$(30) \quad 1 + 2 \sum_{n=1}^{\infty} n^{-1} T_n(x) z^n = -\log(1 - 2xz + z^2)$$

$$(31) \quad \sum_{n=0}^{\infty} U_n(x) z^n = (1 - 2xz + z^2)^{-1}$$

$$(32) \quad \sum_{n=0}^{\infty} g_n T_n(x) z^n = 2^{-\frac{1}{2}} R^{-1} (1 - xz + R)^{\frac{1}{2}}$$

$$(33) \quad \sum_{n=0}^{\infty} g_{n+1} U_n(x) z^n = 2^{-\frac{1}{2}} R^{-1} (1 - xz + R)^{-\frac{1}{2}}.$$

In all five formulas

$$-1 < x < 1, \quad |z| < 1.$$

In the last two formulas

$$R = (1 - 2xz + z^2)^{\frac{1}{2}}.$$

Equation (31) is a special case of 10.9(29), and (30) is a limiting case of the same relation: (29) can be derived from (30). $R = 1$ and $\log R^2 = 0$ when $z = 0$. Formulas (32) and (33) are special cases of 10.9(28).

(vi) *Integral representations.* Contour integrals which represent Tchebichef polynomials follow from any of the generating functions.

(vii) *Miscellaneous results.*

$$(34) \quad 2 T_m(x) T_n(x) = T_{n+m}(x) - T_{n-m}(x) \quad n \geq m$$

$$(35) \quad 2(x^2 - 1) U_{n-1}(x) U_{n-1}(x) = T_{n+m}(x) - T_{n-m}(x) \quad n \geq m$$

$$(36) \quad 2 T_m(x) U_{n-1}(x) = U_{n+m-1}(x) + U_{n-m-1}(x) \quad n > m$$

$$(37) \quad 2 T_n(x) U_{n-1}(x) = U_{n+m-1}(x) - U_{n-m-1}(x) \quad n > m$$

$$(38) \quad 2[T_n(x)]^2 = 1 + 2 T_{2n}(x), \quad 2 T_n(x) U_{n-1}(x) = U_{2n-1}(x)$$

$$(39) \quad 2(1 - x^2) [U_{n-1}(x)]^2 = 1 - 2 T_{2n}(x)$$

$$(40) \quad \sum_{n=0}^N T_{2n}(x) = \frac{1}{2} + \frac{1}{2} U_{2N}(x), \quad \sum_{n=0}^{N-1} T_{2n+1}(x) = \frac{1}{2} U_{2N-1}(x)$$

$$(41) \quad 2(1 - x^2) \sum_{n=0}^N U_{2n}(x) = 1 - T_{2N+2}(x)$$

$$(42) \quad 2(1 - x^2) \sum_{n=0}^{N-1} U_{2n+1}(x) = x - T_{2N+1}(x).$$

All these formulas are paraphrases of trigonometric identities.

Mehler's integral 10.10(43) may be interpreted as a relation between Legendre and Tchebichef polynomials. Inverting this relationship, Tricomi (1935) found

$$(43) \quad (n + \frac{1}{2})(1 + x)^{\frac{1}{2}} \int_{-1}^x (x - t)^{-\frac{1}{2}} P_n(t) dt = T_n(x) + T_{n+1}(x)$$

$$(44) \quad (n + \frac{1}{2})(1 - x)^{\frac{1}{2}} \int_x^1 (t - x)^{-\frac{1}{2}} P_n(t) dt = T_n(x) - T_{n+1}(x).$$

From 10.9(21) and 10.9(22) we obtain

$$(45) \quad P_n^{(\frac{1}{2}, -\frac{1}{2})}(2x^2 - 1) = g_n U_{2n}(x)$$

$$(46) \quad x P_n^{(-\frac{1}{2}, \frac{1}{2})}(2x^2 - 1) = g_n T_{2n+1}(x).$$

Finally we note the principal value integrals

$$(47) \quad \oint_{-1}^1 (y - x)^{-1} (1 - y^2)^{-\frac{1}{2}} T_n(y) dy = \pi U_{n-1}(x)$$

$$(48) \quad \oint_{-1}^1 (y - x)^{-1} (1 - y^2)^{\frac{1}{2}} U_{n-1}(y) dy = -\pi T_n(x) \quad n = 1, 2, \dots$$

which are paraphrases of trigonometric integrals and are of importance in the theory of the integral equation sometimes called the airfoil equation.

10.12. Laguerre polynomials

The polynomials $L_n^\alpha(x)$ are the suitably standardized orthogonal polynomials associated with

$$(1) \quad a = 0, \quad b = \infty, \quad w(x) = e^{-x} x^a, \quad X = x \quad a > -1.$$

Instead of $L_n^0(x)$ it is usual to write $L_n(x)$. This is the polynomial introduced by Laguerre. The $L_n^\alpha(x)$ are often called generalized Laguerre polynomials, but we shall call them Laguerre polynomials simply. Equivalent polynomials have also been discussed by Sonine (1880, p. 41).

(i) *Standardization.* We shall adopt the standardization $k_n = (-1)^n/n!$. The standardizations $k_n = (-1)^n$ and, less frequently, $k_n = 1$ are sometimes used.

(ii) *Constants.*

$$(2) \quad n! h_n = \Gamma(a + n + 1), \quad n! k_n = (-1)^n,$$

$$nr_n = -(n + a), \quad K_n = n!$$

$$(3) \quad (n + 1) A_n = -1, \quad (n + 1) B_n = 2n + a + 1, \quad (n + 1) C_n = n + a$$

$$(4) \quad \lambda_n = n, \quad \alpha_n = n, \quad \beta_n = -(n + a).$$

(iii) *Relationships.*

$$(5) \quad n! L_n^\alpha(x) = e^x x^{-a} D^n(e^{-x} x^{n+a})$$

$$(6) \quad L_0^\alpha(x) = 1, \quad L_1^\alpha(x) = a + 1 - x$$

$$(7) \quad L_n^\alpha(x) = \sum_{m=0}^n \binom{n+a}{n-m} \frac{(-x)^m}{m!}$$

$$(8) \quad (n + 1) L_{n+1}^\alpha(x) - (2n + a + 1 - x) L_n^\alpha(x) + (n + a) L_{n-1}^\alpha(x) = 0$$

$$(9) \quad \sum_{m=0}^n \frac{m!}{\Gamma(m + a + 1)} L_m^\alpha(x) L_m^\alpha(y) \\ = \frac{(n + 1)!}{\Gamma(n + a + 1)} \frac{1}{x - y} [L_n^\alpha(x) L_{n+1}^\alpha(y) - L_{n+1}^\alpha(x) L_n^\alpha(y)]$$

$$(10) \quad xy'' + (a + 1 - x)y' + ny = 0, \quad y = L_n^\alpha(x)$$

$$(11) \quad (xz')' + \left(n + \frac{a+1}{2} - \frac{x}{4} - \frac{a^2}{4x} \right) z = 0, \quad z = e^{-\frac{1}{2}x} x^{\frac{1}{2}a} L_n^\alpha(x)$$

$$\begin{aligned}
 (12) \quad x \frac{d}{dx} L_n^a(x) &= n L_n^a(x) - (n+a) L_{n-1}^a(x) \\
 &= (n+1) L_{n+1}^a(x) - (n+a+1-x) L_n^a(x)
 \end{aligned}$$

$$(13) \quad L_n^a(0) = \binom{n+a}{n} = \frac{(a+1)_n}{n!}.$$

(iv) *Hypergeometric functions.* Laguerre polynomials are connected with the confluent hypergeometric functions of Chapter VI. From the explicit representation (7)

$$\begin{aligned}
 (14) \quad L_n^a(x) &= \binom{n+a}{n} \Phi(-n, a+1; x) \\
 &= \frac{(-1)^n}{n!} \Psi(-n-a, 1-a; x).
 \end{aligned}$$

From this we have

$$(15) \quad \frac{d}{dx} L_n^a(x) = -L_{n-1}^{a+1}(x)$$

confirming statement (i) of sec. 10.6,

$$(16) \quad \frac{d}{dx} [L_n^a(x) - L_{n+1}^a(x)] = L_n^a(x),$$

and many other formulas which are instances of relations between contiguous confluent hypergeometric functions.

The general solution of Laguerre's differential equation (10) may be obtained from the theory of confluent hypergeometric functions.

(v) *Generating functions.*

$$(17) \quad \sum_{n=0}^{\infty} L_n^a(x) z^n = (1-z)^{-a-1} \exp \frac{xz}{z-1} \quad |z| < 1$$

$$(18) \quad \sum_{n=0}^{\infty} [\Gamma(n+a+1)]^{-1} L_n^a(x) z^n = (xz)^{-\frac{1}{2}a} e^z J_a[2(xz)^{\frac{1}{2}}]$$

$$(19) \quad \sum_{n=0}^{\infty} L_n^{a-n}(x) z^n = e^{-xz} (1+z)^a \quad |z| < 1$$

$$\begin{aligned}
 (20) \quad \sum_{n=0}^{\infty} \frac{n!}{\Gamma(n+a+1)} L_n^a(x) L_n^a(y) z^n \\
 = (1-z)^{-1} \exp \left(-z \frac{x+y}{1-z} \right) (xyz)^{-\frac{1}{2}a} I_a \left[2 \frac{(xyz)^{\frac{1}{2}}}{1-z} \right] \quad |z| < 1.
 \end{aligned}$$

The function on the right-hand side of (17) is the most common generating function and may be established by means of (7). Equation (18) is due to Doetsch and follows from (17) by means of the Laplace transformation. Equation (19) follows from (7) and is due to Erdélyi. Equation (20) is a bilinear generating function and is known as the Hille-Hardy formula (see also Myller-Lebedeff, 1907).

(vi) *Integral representations.* Contour integrals representing Laguerre polynomials may be obtained from the Rodrigues formula (5) and from any of the generating functions in an obvious manner. In addition, the connection (14) with confluent hypergeometric functions may be exploited (cf. sec. 6.11). We mention only the following integrals

$$(21) \quad n! L_n^\alpha(x) = e^x x^{-\frac{1}{2}\alpha} \int_0^\infty e^{-t} t^{n+\frac{1}{2}\alpha} J_\alpha[2(tx)^{\frac{1}{2}}] dt$$

$$(22) \quad 2\pi i 2^\alpha L_n^\alpha(x) = (-1)^n e^{\frac{1}{2}x} \int_{-1}^{+1} e^{-\frac{1}{2}xz} \left(\frac{1+z}{1-z} \right)^k (1-z^2)^{\frac{1}{2}\alpha-1} dz$$

$$k = n + \frac{1}{2}\alpha + \frac{1}{2}.$$

The first of these is a consequence of 6.11(5); and the second is due to Tricomi.

(vii) *Miscellaneous results.* The number of results under this heading is tremendous, and many of them were discovered several times. We give a small selection only and do not attempt to credit the formulas to their original discoverers.

Contiguous polynomials. In addition to the recurrence formula (8) we have

$$(23) \quad x L_n^{\alpha+1}(x) = (n + \alpha + 1) L_n^\alpha(x) - (n + 1) L_{n+1}^\alpha(x)$$

$$= (n + \alpha) L_{n-1}^\alpha(x) - (n - x) L_n^\alpha(x)$$

$$(24) \quad L_n^{\alpha-1}(x) = L_n^\alpha(x) - L_{n-1}^\alpha(x)$$

$$(25) \quad (n + \alpha) L_n^{\alpha-1}(x) = (n + 1) L_{n+1}^\alpha(x) - (n + 1 - x) L_n^\alpha(x).$$

Differentiation formulas and indefinite integrals. In addition to (5) and (12) we have

$$(26) \quad D^n [x^{-\alpha-1} \exp(-x^{-1})] = (-1)^n n! x^{-\alpha-n-1} L_n^\alpha(x^{-1}) \exp(-x^{-1})$$

$$(27) \quad D^m [x^\alpha L_n^\alpha(x)] = (n - m + \alpha + 1)_m x^{\alpha-m} L_n^{\alpha-m}(x)$$

$$(28) \quad n! D^m [e^{-x} x^\alpha L_n^\alpha(x)] = (m + n)! e^{-x} x^{\alpha-m} L_{m+n}^{\alpha-m}(x)$$

$$(29) \int_x^\infty e^{-y} L_n^\alpha(y) dy = e^{-x} [L_n^\alpha(x) - L_{n-1}^\alpha(x)]$$

$$(30) \Gamma(a + \beta + n + 1) \int_0^x (x - y)^{\beta-1} y^a L_n^\alpha(y) dy \\ = \Gamma(a + n + 1) \Gamma(\beta) x^{a+\beta} L_n^{a+\beta}(x) \quad \text{Re } a > -1, \quad \text{Re } \beta > 0$$

$$(31) \int_0^x L_m(y) L_n(x - y) dy = \int_0^x L_{m+n}(y) dy = L_{m+n}(x) - L_{m+n+1}(x).$$

Further indefinite integrals follow from the product theorem of Laplace transforms.

Laplace integrals. With the notation

$$\mathcal{L}[F(t)] = \int_0^\infty e^{-st} F(t) dt$$

we have

$$(32) \mathcal{L}[t^\alpha L_n^\alpha(t)] = \frac{\Gamma(a + n + 1)(s - 1)^n}{n! s^{a+n+1}} \quad \text{Re } a > -1, \quad \text{Re } s > 0$$

$$(33) n! \Gamma(a + 1) \mathcal{L}[t^\beta L_n^\alpha(t)] \\ = \Gamma(\beta + 1) \Gamma(a + n + 1) s^{-\beta-1} F(-n, \beta + 1; a + 1; s^{-1}) \\ \text{Re } \beta > -1, \quad \text{Re } s > 0$$

$$(34) \mathcal{L}[t^{\frac{1}{2}a+n} J_a(2(kt)^{\frac{1}{2}})] = n! k^{\frac{1}{2}a} s^{-a-n-1} e^{-k/s} L_n^\alpha(k/s).$$

Limit formulas.

$$(35) L_n^\alpha(x) = \lim_{\beta \rightarrow \infty} P_n^{(\alpha, \beta)}(1 - 2x/\beta)$$

$$(36) \lim_{n \rightarrow \infty} [n^{-\alpha} L_n^\alpha(x/n)] = x^{-\frac{1}{2}\alpha} J_\alpha(2x^{\frac{1}{2}}).$$

Finite difference formula. With the notation

$$\Delta_a f(a) = f(a + 1) - f(a), \quad \Delta_a^{n+1} f(a) = \Delta_a(\Delta_a^n f(a)) \quad n = 1, 2, \dots$$

we have

$$\Delta_a^n f(a) = \sum_{m=0}^n (-1)^{n-m} \binom{n}{m} f(a + m) \quad n = 1, 2, \dots$$

and hence

$$(37) L_n^\alpha(x) = (-1)^n \frac{\Gamma(a + n + 1)}{n! x^a} \Delta_a^n \frac{x^a}{\Gamma(a + 1)}.$$

Finite sums. In addition to those already recorded we have

$$(38) \quad \sum_{n=0}^{\infty} L_n^{\alpha}(x) = L_n^{\alpha+1}(x) = x^{-1} [(x-n) L_n^{\alpha}(x) + (n+\alpha) L_{n-1}^{\alpha}(x)]$$

$$(39) \quad L_n^{\alpha}(x) = \sum_{m=0}^n (m!)^{-1} (n-m)! L_{n-m}^{\beta}(x)$$

$$(40) \quad L_n^{\alpha}(\lambda x) = \sum_{m=0}^n \binom{n+\alpha}{m} \lambda^{n-m} (1-\lambda)^m L_{n-m}^{\alpha}(x)$$

$$(41) \quad \sum_{n=0}^{\infty} L_n^{\alpha}(x) L_{n-m}^{\beta}(y) = L_n^{\alpha+\beta+1}(x+y)$$

$$(42) \quad n! L_n^{\alpha}(x) L_n^{\alpha}(y)$$

$$= \Gamma(\alpha+n+1) \sum_{m=0}^n [m! \Gamma(\alpha+m+1)]^{-1} (xy)^m L_{n-m}^{\alpha+2m}(x+y).$$

Infinite series: generating functions have already been given [(17) to (20)], Bessel function expansions are in sec. 10.15, and other examples of infinite series involving Laguerre polynomials are in sec. 10.20.

10.13. Hermite polynomials

Hermite polynomials are orthogonal polynomials associated with the interval $(-\infty, \infty)$ and an exponential weight function. Unfortunately, the notations adopted by different authors vary a great deal. The simplest form of the exponential weight function seems to be $\exp(-x^2)$, but for applications in mathematical statistics $\exp(-\frac{1}{2}x^2)$ is preferable. Of the most important books on the subject, Courant-Hilbert, Doetsch, Sansone, Szegő use $\exp(-x^2)$, and Appell and Kampé de Fériet, Jahnke-Emde, Magnus-Oberhettinger, Pólya-Szegő, and Tricomi use the weight function $\exp(-\frac{1}{2}x^2)$. In this chapter we shall adopt Szegő's (1939) notation and regard Hermite polynomials, $H_n(x)$, as the suitably standardized orthogonal polynomials associated with

$$(1) \quad a = -\infty, \quad b = \infty, \quad w(x) = \exp(-x^2), \quad X = 1.$$

The orthogonal polynomials associated with the weight function $\exp(-\frac{1}{2}x^2)$ may be denoted by $He_n(x)$. These polynomials may also be expressed in terms of parabolic cylinder functions [see 8.2(9)].

(i) *Standardization.* We shall adopt the standardization $K_n = (-1)^n$. This agrees with the standardization adopted, among other authors, by Courant-Hilbert, Feldheim, Hille, and Szegő. The standardization $K_n = 1$ is used by Doetsch, Erdélyi, Sansone, and others.

Hermite polynomials, so standardized, have been expressed by Szegő and Koschmieder in terms of Laguerre polynomials.

$$(2) \quad H_{2m}(x) = (-1)^m 2^{2m} m! L_m^{-1/2}(x^2)$$

$$(3) \quad H_{2m+1}(x) = (-1)^m 2^{2m+1} m! x L_m^{1/2}(x^2).$$

These expressions show that $H_n(x)$ is an even or an odd function of x according as n is even or odd. These formulas are analogous to (and are in fact limiting cases of) 10.9(21) and 10.9(22).

(ii) *Constants.*

$$(4) \quad h_n = \pi^{1/2} 2^n n!, \quad k_n = 2^n, \quad r_n = 0$$

$$(5) \quad K_n = (-1)^n, \quad A_n = 2, \quad B_n = 0, \quad C_n = 2n$$

$$(6) \quad \lambda_n = 2n, \quad a_n = 0, \quad \beta_n = 2n.$$

(iii) *Relationships.*

$$(7) \quad H_n(x) = (-1)^n e^{x^2} D^n e^{-x^2}$$

$$(8) \quad H_0(x) = 1, \quad H_1(x) = 2x$$

$$(9) \quad H_n(x) = n! \sum_{m=0}^{[n/2]} \frac{(-1)^m (2x)^{n-2m}}{m! (n-2m)!}.$$

Here $[n/2]$ is $n/2$ or $(n-1)/2$ according as n is even or odd.

$$(10) \quad H_{n+1}(x) - 2x H_n(x) + 2n H_{n-1}(x) = 0$$

$$(11) \quad \sum_{m=0}^n \frac{H_m(x) H_m(y)}{2^m m!} = \frac{H_{n+1}(x) H_n(y) - H_n(x) H_{n+1}(y)}{2^{n+1} n! (x-y)}$$

$$(12) \quad y'' - 2xy' + 2ny = 0, \quad y = H_n(x)$$

$$(13) \quad z'' + (2n+1-x^2)z = 0, \quad z = \exp(-\frac{1}{2}x^2) H_n(x)$$

$$(14) \quad H_n(-x) = (-1)^n H_n(x), \quad H'_n(x) = 2n H_{n-1}(x)$$

$$(15) \quad H_{2m}(0) = (-1)^m (2m)!/m!, \quad H_{2m+1}(0) = 0.$$

(iv) *Hypergeometric functions.* Hermite polynomials are connected with parabolic cylinder functions which are special confluent hypergeometric functions

$$(16) \quad H_n(x) = 2^{\frac{1}{2}n} \exp(\frac{1}{2}x^2) D_n(2^{\frac{1}{2}}x) = 2^n \Psi(-\frac{1}{2}n, \frac{1}{2}; x^2)$$

$$(17) \quad m! H_{2m}(x) = (-1)^m (2m)! \Phi(-m, \frac{1}{2}; x^2)$$

$$(18) \quad m! H_{2m+1}(x) = (-1)^m (2m+1)! 2x \Phi(-m, 3/2; x^2).$$

The general solution of Hermite's differential equation (12), or of the self-adjoint form (13) (which is virtually Weber's equation), may be obtained from the theory of parabolic cylinder functions.

(v) *Generating functions.*

$$(19) \quad \sum_{n=0}^{\infty} H_n(x) z^n / n! = \exp(2xz - z^2)$$

$$(20) \quad \sum_{m=0}^{\infty} (-1)^m H_{2m}(x) z^{2m} / (2m)! = \exp(z^2) \cos(2^{\frac{1}{2}}xz)$$

$$(21) \quad \sum_{m=0}^{\infty} (-1)^m H_{2m+1}(x) z^{2m+1} / (2m+1)! = \exp(z^2) \sin(2^{\frac{1}{2}}xz)$$

$$(22) \quad \sum_{n=0}^{\infty} \frac{(\frac{1}{2}z)^n}{n!} H_n(x) H_n(y) = (1-z^2)^{-\frac{1}{2}} \exp \left\{ \frac{2xyz - (x^2 + y^2)z^2}{1-z^2} \right\}.$$

Equation (19) is the well-known generating function, (20) and (21) may be derived from (19), and (22) is Mehler's formula.

(vi) *Integral representations.* Contour integrals follow in the usual manner from (7) or from any of the generating functions. In addition, the connection with parabolic cylinder functions may be exploited (see sec. 8.3). We note explicitly

$$(23) \quad e^{-x^2} H_n(x) = 2^{n+1} \pi^{-\frac{1}{2}} \int_0^{\infty} e^{-t^2} t^n \cos(2xt - \frac{1}{2}n\pi) dt.$$

(vii) *Miscellaneous results.* See remarks under 10.12(vii).

Limits.

$$(24) \quad \lim_{m \rightarrow \infty} \left[\frac{(-1)^m m^{\frac{1}{2}}}{2^{2m} m!} H_{2m} \left(\frac{x}{2m^{\frac{1}{2}}} \right) \right] = \pi^{-\frac{1}{2}} \cos x$$

$$(25) \quad \lim_{m \rightarrow \infty} \left[\frac{(-1)^m}{2^{2m} m!} H_{2m+1} \left(\frac{x}{2m^{\frac{1}{2}}} \right) \right] = 2\pi^{-\frac{1}{2}} \sin x.$$

Integrals.

$$(26) \quad \int_0^x e^{-y^2} H_n(y) dy = H_{n-1}(0) - e^{-x^2} H_{n-1}(x)$$

$$(27) \quad \int_n^x H_n(y) dy = [2(n+1)]^{-1} [H_{n+1}(x) - H_{n+1}(0)]$$

$$(28) \int_{-\infty}^{\infty} e^{-y^2} H_{2m}(xy) dy = \pi^{1/2} \frac{(2m)!}{m!} (x^2 - 1)^m$$

$$\int_{-\infty}^{\infty} e^{-y^2} y H_{2m+1}(xy) dy = \pi^{1/2} \frac{(2m+1)!}{m!} x (x^2 - 1)^m$$

$$(29) \int_{-\infty}^{\infty} e^{-y^2} y^n H_n(xy) dy = \pi^{1/2} n! P_n(x).$$

Here $P_n(x)$ is the Legendre polynomial.

Gauss transforms.

$$G_x^u[F(y)] = (2\pi u)^{-1/2} \int_{-\infty}^{\infty} F(y) \exp[-(x-y)^2/(2u)] dy$$

is the Gauss transform (with parameter u). We have

$$(30) G_x^u[H_n(y)] = (1-2u)^{1/2n} H_n[(1-2u)^{-1/2}x] \quad 0 \leq u < 1/2$$

$$(31) G_x^{1/2}[H_n(y)] = (2x)^n, \quad G_x^{1/2}[y^n] = (2i)^{-n} H_n(ix).$$

Connection with Laguerre polynomials. In addition to (2) and (3) we have

$$(32) \sum_{k=0}^n \binom{n}{k} H_{2k}(x) H_{2n-2k}(y) = (-1)^n n! L_n(x^2 + y^2)$$

$$(33) \int_0^{\infty} e^{-y^2} [H_n(y)]^2 \cos(2^{1/2} xy) dy = \pi^{1/2} 2^{n-1} n! L_n(x^2)$$

$$(34) \Gamma(n+a+1) \int_{-1}^1 (1-t^2)^{a-1/2} H_{2n}(x^{1/2}t) dt = (-1)^n \pi^{1/2} (2n)! \Gamma(a+1/2) L_n^a(x)$$

$$\operatorname{Re} a > -1/2.$$

The first two of these formulas are due to Feldheim, the last one to Uspensky (1927).

Finite sums. In addition to those already given we have, among others, the following relations.

$$(35) \sum_{m=0}^n (2^m m!)^{-1} [H_m(x)]^2 = (2^{n+1} n!)^{-1} \{[H_{n+1}(x)]^2 - H_n(x) H_{n+2}(x)\}$$

$$(36) \sum_{k=0}^{\min(m,n)} (-2)^k k! \binom{m}{k} \binom{n}{k} H_{m-k}(x) H_{n-k}(x) = H_{m+n}(x)$$

$$(37) \sum_{k=0}^{\min(m,n)} 2^k k! \binom{m}{k} \binom{n}{k} H_{m+n-2k}(x) = H_m(x) H_n(x)$$

$$(38) \sum_{k=0}^m \binom{m}{k} H_k(2^{1/2}x) H_{m-k}(2^{1/2}y) = 2^{1/2m} H_m(x+y)$$

$$(39) \quad \sum_{k=0}^n \binom{2n}{2k} H_{2k}(2^{\frac{1}{2}}x) H_{2n-2k}(2^{\frac{1}{2}}y) \\ = 2^{n-1} [H_{2n}(x+y) + H_{2n}(x-y)]$$

$$(40) \quad \sum_{m_1 + \dots + m_r = n} \frac{a_1^{m_1}}{m_1!} \dots \frac{a_r^{m_r}}{m_r!} H_{m_1}(x_1) \dots H_{m_r}(x_r) \\ = \frac{(a_1^2 + \dots + a_r^2)^{\frac{1}{2}n}}{n!} H_n \left[\frac{a_1 x_1 + \dots + a_r x_r}{(a_1^2 + \dots + a_r^2)^{\frac{1}{2}}} \right].$$

Equation (35) is due to Demir and Hsü. The last three formulas are addition theorems and can easily be proved by means of the generating function (19). In (40) the sum is extended over all non-negative integers $m_1, \dots, m_r$ whose sum is n .

Infinite series. Generating functions have already been given [(19) to (22)]. For expansions in spherical Bessel functions see sec. 10.15, and for other infinite series involving Hermite polynomials see sec. 10.20.

10.14. Asymptotic behavior of Jacobi, Gegenbauer and Legendre polynomials

The behavior of Jacobi polynomials as $n \rightarrow \infty$ and at the same time $x \rightarrow 1$ in a suitable manner, is given by 10.8(41). The corresponding behavior as $x \rightarrow -1$ follows from 10.8(13), and the behavior of Gegenbauer and Legendre polynomials may be obtained by means of 10.9(4) and 10.10(3). The behavior of Jacobi polynomials as $\beta \rightarrow \infty$ and $x \rightarrow 1$ in a suitable manner is given by 10.12(35).

For the investigation of the convergence of infinite series of Jacobi polynomials, and for many other purposes, it is desirable to determine the behavior of Jacobi polynomials for fixed α, β, x , and for $n \rightarrow \infty$. The example of Tchebichef polynomials

$$T_n(\cos \theta) = \cos n \theta, \quad x = \cos \theta$$

suggests that the asymptotic behavior will be different according as x is on the interval $(-1, 1)$ (θ real) or outside it (θ complex). Also the end-points of the interval have to be considered separately. In the present section we shall entirely omit the case in which x is outside $(-1, 1)$ and refer the reader in this respect to Szegő (1939, Chapter VIII). We shall give certain results for $-1 < x < 1$; and all estimates given in this section hold uniformly in any interval $-1 + \epsilon \leq x \leq 1 - \epsilon$ ($\epsilon > 0$). We shall also give important results for the case that x is in the neighborhood of ± 1 .

Proofs of the results to be given are based either on explicit series or integral representations (with integral representations the method of steepest descent is frequently used), on generating functions (Darboux's method), or else on the differential equations (Liouville's method and its later developments).

Darboux has proved (from the generating function)

$$(1) \quad P_n(\cos \theta) = 2g_n \sum_{m=0}^{M-1} \frac{g_m (\frac{1}{2})_m}{(n-m+\frac{1}{2})_m} \frac{\cos[(n-m+\frac{1}{2})\theta - (\frac{1}{2}m + \frac{1}{4})\pi]}{(2 \sin \theta)^{m+\frac{1}{2}}} + O(n^{-M-\frac{1}{2}}) \quad 0 < \theta < \pi$$

where g_n is defined by 10.10(21).

A similar formula was obtained by Stieltjes in a manner permitting to estimate the remainder.

$$(2) \quad P_n(\cos \theta) = \frac{2}{\pi} \sum_{m=0}^{M-1} \frac{n! g_m}{(m+\frac{1}{2})_{n+1}} \frac{\cos[(n+m+\frac{1}{2})\theta - (\frac{1}{2}m + \frac{1}{4})\pi]}{(2 \sin \theta)^{m+\frac{1}{2}}} + R_M(\theta) \quad 0 < \theta < \pi$$

where

$$(3) \quad |R_M(\theta)| \leq \frac{2}{\pi} \frac{n! g_M}{(M+\frac{1}{2})_{n+1}} \frac{A}{(2 \sin \theta)^{M+\frac{1}{2}}}$$

and

$$(4) \quad A = 2 \sin \theta \quad \text{if} \quad \sin^2 \theta \geq \frac{1}{2}$$

$$A = |\cos \theta|^{-1} \quad \text{if} \quad \sin^2 \theta \leq \frac{1}{2}$$

so that in any event $1 \leq A \leq 2$.

If $2 \sin \theta > 1$, i.e., for $\pi/6 < \theta < 5\pi/6$, we may make $M \rightarrow \infty$ in both (1) and (2) and obtain convergent trigonometric expansions of Legendre polynomials.

For the neighborhood of $x = 1$ we have Hilb's formula

$$(5) \quad P_n(\cos \theta) = (\theta \csc \theta)^{\frac{1}{2}} J_0[(n+\frac{1}{2})\theta] + O(n^{-3/2})$$

valid uniformly for $0 \leq \theta \leq \pi - \epsilon$ ($\epsilon > 0$). For more precise bounds for the error term see Szegő (1939, p. 189). For an expansion of Legendre polynomials in series of Bessel functions see Szegő (1933). If 10.14(11) is specialized for $\alpha = \beta = 0$ and the confluent hypergeometric function there is expanded in a series of Bessel functions by means of 6.12(6), the result is

$$(6) \quad P_n(x) = [4/(x+3)]^{n+1} e^{-\xi/(2n+1)} [J_0(2\xi^{1/2}) + \frac{\xi}{8n^2} J_2(2\xi^{1/2}) + O(n^{-3})]$$

where

$$2(x+3)\xi = (1-x)(2n+1)^2.$$

Some of these results can be extended to Gegenbauer polynomials and to a lesser extent also to Jacobi polynomials.

$$(7) \quad C_n^\lambda(\cos \theta) = 2 \frac{(\lambda)_n}{n!} \sum_{m=0}^{M-1} \frac{(\lambda)_m (1-\lambda)_m}{(n-m+\lambda)_m m!} \\ \times \frac{\cos[(n-m+\lambda)\theta - \frac{1}{2}(m+\lambda)\pi]}{(2 \sin \theta)^{\lambda+m}} + O(n^{-M-\frac{1}{2}}) \\ \lambda \neq 0, -1, -2, \dots, \quad 0 < \theta < \pi$$

$$(8) \quad C_n^\lambda(\cos \theta) = 2 \frac{\Gamma(2\lambda+n)}{[\Gamma(\lambda)]^2} \sum_{m=0}^{M-1} \frac{(1-\lambda)_m}{(m+\lambda)_{n+1} m!} \\ \times \frac{\cos[(n+m+\lambda)\theta - \frac{1}{2}(m+\lambda)\pi]}{(2 \sin \theta)^{\lambda+m}} + R_M(\theta) \quad 0 < \lambda < 1, \quad 0 < \theta < \pi$$

$$(9) \quad |R_M(\theta)| \leq 2 \frac{\Gamma(n+2\lambda)}{[\Gamma(\lambda)]^2} \frac{(1-\lambda)_M}{(M+\lambda)_{n+1} M!} \frac{A}{(2 \sin \theta)^{\lambda+M}}$$

and A is given by (4).

$$(10) \quad P_n^{(\alpha, \beta)}(\cos \theta) = \frac{\cos\{[n + \frac{1}{2}(\alpha + \beta + 1)]\theta - (\frac{1}{2}\alpha + \frac{1}{4})\pi\}}{(\pi n)^{1/2} (\sin \frac{1}{2}\theta)^{\alpha+1/2} (\cos \frac{1}{2}\theta)^{\beta+1/2}} + O(n^{-3/2})$$

$$\alpha, \beta \text{ real}, \quad 0 < \theta < \pi.$$

A formula of Hilb's type for Jacobi polynomials has been given by Szegő and by Rau: see Szegő (1939, p. 191). Tricomi (1950a) obtained the expansion

$$(11) \quad P_n^{(\alpha, \beta)}(x) = e^{-z} \left(\frac{1+x}{2} \right)^{-N} \sum_{m=0}^{\infty} \frac{\Gamma(N+m) \Gamma(\alpha+n+1)}{n! \Gamma(N) \Gamma(\alpha+m+1)} \\ \times A_m(k, \frac{1}{2}\alpha + \frac{1}{2}) [z/(2k)]^m \Phi(-n-\beta, \alpha+m+1; z)$$

where

$$(12) \quad k = n + \frac{1}{2}\alpha + \frac{1}{2}, \quad N = n + \alpha + \beta + 1, \quad x = 1 - 4z/(2k+z) \quad |z| < 2|k|,$$

Φ is the confluent hypergeometric series and the A_m are the coefficients defined in sec. 6.12. Using the expansion 6.12(6), a Bessel function expansion may be obtained for Jacobi polynomials. In the particular case $\alpha = \beta = 0$ it leads to (6).

10.15. Asymptotic behavior of Laguerre and Hermite polynomials

The general remarks of the preceding section apply here too, but the situation is more involved on account of the infinite interval. The polynomials are oscillatory in part of the interval, and monotonic outside this part.

The asymptotic behavior of Laguerre and Hermite polynomials as $n \rightarrow \infty$ and at the same time $x \rightarrow 0$ in a suitable manner is given by 10.12 (36), 10.13 (24), and 10.13 (25).

For real α and fixed $x > 0$, or uniformly in $0 < \epsilon \leq x \leq \omega < \infty$, we have Fejér's formula

$$(1) \quad L_n^\alpha(x) = \pi^{-1/2} e^{1/2x} x^{-1/2\alpha-1/4} n^{1/2\alpha-1/4} \cos[2(nx)^{1/2} - 1/2\alpha\pi - 1/4\pi] + O(n^{1/2\alpha-3/4})$$

which has been generalized by Perron (see Szegő, 1939, p. 192). Sansone (1950) gave a two-term approximation with an estimate of the error. This formula fails for small x , but there is a formula of Hilb's type

$$(2) \quad e^{-1/2x} x^{1/2\alpha} L_n^\alpha(x) = \frac{\Gamma(n+\alpha+1)}{(1/4\nu)^{1/2\alpha} n!} J_\alpha[(\nu x)^{1/2}] + O(n^{1/2\alpha-3/4})$$

valid for $\alpha > -1$ uniformly in $0 < x \leq \omega < \infty$. The notation

$$(3) \quad \nu = 4n + 2\alpha + 2$$

is used in (2), and will be retained throughout this section.

The behavior of Laguerre polynomials when $n \rightarrow \infty$ and x is unrestricted has been investigated by several authors (see sec. 6.13). We shall confine ourselves to a brief survey here, based on a memoir by Tricomi (1949). Tricomi distinguishes four cases according as x is near 0, in the oscillatory region, near ν , or in the monotonic region.

The expansion

$$(4) \quad n! e^{-1/2x} L_n^\alpha(x) = \Gamma(\alpha+n+1)(\nu x/4)^{-1/2\alpha} \sum_{m=0}^{\infty} A_m^*(x/\nu)^{1/2m} J_{\alpha+m}([\nu x]^{1/2})$$

which is a particular case of 6.12(11) and in which

$$(5) \quad A_0^* = 1, \quad A_1^* = 0, \quad A_2^* = 1/2\alpha + 1/2$$

$$(m+2) A_{m+2}^* = (m+\alpha+1) A_m^* - 1/2\nu A_{m-1}^*, \quad m = 1, 2, \dots,$$

converges uniformly in any bounded region of the complex variable x . By considering the order of magnitude of successive terms, one sees that (4) has the character of an asymptotic expansion as $n \rightarrow \infty$ provided that $x = O(n^\lambda)$ with $\lambda < 1/3$. This establishes the behavior of $L_n^\alpha(x)$ "near" the origin.

A similar expansion,

$$(6) \quad n! (ux)^{\frac{1}{2}\alpha} e^{-ux} L_n^\alpha(x) = \Gamma(\alpha + n + 1) \sum_{m=0}^{\infty} A_m^{(\alpha)}(h) (x/u)^{\frac{1}{2}m} J_{\alpha+m}(2[ux]^{\frac{1}{2}})$$

with appropriate coefficients is due to Toscano (1949), in case $u = n$ to Tricomi (1941).

In the oscillatory region, $0 < x < \nu$, Tricomi puts

$$(7) \quad x = \nu \cos^2 \theta, \quad 0 < \theta < \frac{1}{2}\pi, \quad 4\Theta = \nu(2\theta - \sin 2\theta) + \pi$$

and proves that for a fixed θ

$$(8) \quad e^{-\frac{1}{2}x} L_n^\alpha(x) = 2(-1)^n (2\cos \theta)^{-\alpha} (\pi\nu \sin 2\theta)^{-\frac{1}{2}} \\ \times \left[\sum_{m=0}^{M-1} A_m^{(\alpha)}(\theta) (\frac{1}{4}\nu \sin 2\theta)^{-m} \sin(\Theta + 3m\pi/2) + O(n^{-M}) \right]$$

where

$$(9) \quad A_0^{(\alpha)}(\theta) = 1, \quad A_1^{(\alpha)}(\theta) = \frac{1}{12} \left[\frac{5}{4\sin^2 \theta} - (1 - 3\alpha^2) \sin^2 \theta - 1 \right].$$

For the general expression for the $A_m^{(\alpha)}$ see Tricomi (1949).

Near the transition point ν ,

$$(10) \quad e^{-\frac{1}{2}x} L_n^\alpha(x) = \gamma_1 \left\{ A(t) + \left(\frac{4}{3\nu^2} \right)^{1/3} \left[\frac{t^2}{5} A'(t) \right. \right. \\ \left. \left. + \frac{3+5\alpha}{10} \left(t - \frac{\Gamma(1/3)}{2\Gamma(2/3)} \right) A(t) \right] + O(n^{-5/3}) \right\}$$

where

$$(11) \quad t = (4\nu/3)^{-1/3} (\nu - x),$$

$$(12) \quad \pi\gamma_1 = (-1)^n 2^{-\alpha} \left[6^{1/3} \nu^{-1/3} + \frac{3+5\alpha}{10} \frac{\Gamma(1/3)}{\Gamma(2/3)} \nu^{-1} + O(n^{-5/3}) \right],$$

$$(13) \quad A(t) = (\pi/3)(t/3)^{1/2} \{ J_{-1/3}[2(t/3)^{3/2}] + J_{1/3}[2(t/3)^{3/2}] \}$$

is the Airy function, and $A'(t)$ is the derivative of $A(t)$.

Finally, in the monotonic region

$$(14) \quad x = \nu \cosh^2 \theta, \quad \theta > 0, \quad 4\theta = \nu(\sinh 2\theta - 2\theta)$$

$$(15) \quad e^{-\frac{1}{2}x} L_n^\alpha(x) = (-1)^n e^{-\theta} (2 \cosh \theta)^{-\alpha} (\pi \nu \sinh 2\theta)^{-\frac{1}{2}} \\ \times \left[\sum_{m=0}^{n-1} (-1)^m A_m^{[\alpha]}(\theta) \left(\frac{1}{4} \nu \sinh 2\theta\right)^{-m} + O(n^{-N}) \right]$$

where

$$(16) \quad A_0^{[\alpha]}(\theta) = 1, \quad A_1^{[\alpha]}(\theta) = \frac{1}{12} \left[\frac{5}{4 \sinh^2 \theta} - (1 - 3\alpha^2) \sinh^2 \theta + 1 \right].$$

In the following summary of the corresponding results for Hermite polynomials we use the abbreviations

$$(17) \quad N = 2n + 1, \quad m = \begin{cases} \frac{1}{2}n & \text{if } n \text{ is even,} \\ \frac{1}{2}n - \frac{1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

For a fixed real x (or uniformly in any bounded interval)

$$(18) \quad \Gamma(\tfrac{1}{2}n + 1) \exp(-\tfrac{1}{2}x^2) H_n(x) \\ = \Gamma(n + 1) [\cos(N^{\frac{1}{2}}x - \tfrac{1}{2}n\pi) + O(n^{-\frac{1}{2}})].$$

Szegö (1939, p. 194) gives a second term explicitly, and also the general form of the asymptotic expansion.

For the behavior of Hermite polynomials for $n \rightarrow \infty$ and unrestricted x we have the Plancherel-Rotach formulas (Szegö 1939, p. 195). Tricomi's work covers this case too if use is made of 10.13(2) and 10.13(3). The Bessel functions involved in (4) when $\alpha = \pm \frac{1}{2}$ are so-called spherical Bessel functions and can be expressed in closed form. They serve as asymptotic expansions provided that for some $\lambda < 1/3$ the quantity $n^{-\lambda}x$ is bounded as $n \rightarrow \infty$.

The oscillatory region is $0 < |x| < 2m^{\frac{1}{2}}$, and here the expansion (8), with $\alpha = \pm \frac{1}{2}$, may be used. In the neighborhood of the transition points $x = \pm 2m^{\frac{1}{2}}$ we have (10), and in the monotonic region $|x| > 2m^{\frac{1}{2}}$ we have (15).

The basic expansions in series of spherical Bessel functions are particular cases of the more general expansions given by Tricomi (1941):

$$(19) \quad e^{-hx^2} H_{2m}(x) = (-1)^m 2^{2m+1} \left(\frac{1}{2}\right)_m x^2 \sum_{r=0}^{\infty} (2m)^{1-r} C_r' J_{r-1}(2m^{\frac{1}{2}}x)$$

$$(20) \quad e^{-hx^2} H_{2m+1}(x) = (-1)^m 2^{2m+1} (3/2)_m x \sum_{r=0}^{\infty} (2m)^{-r} C_r'' J_r(2m^{\frac{1}{2}}x)$$

where

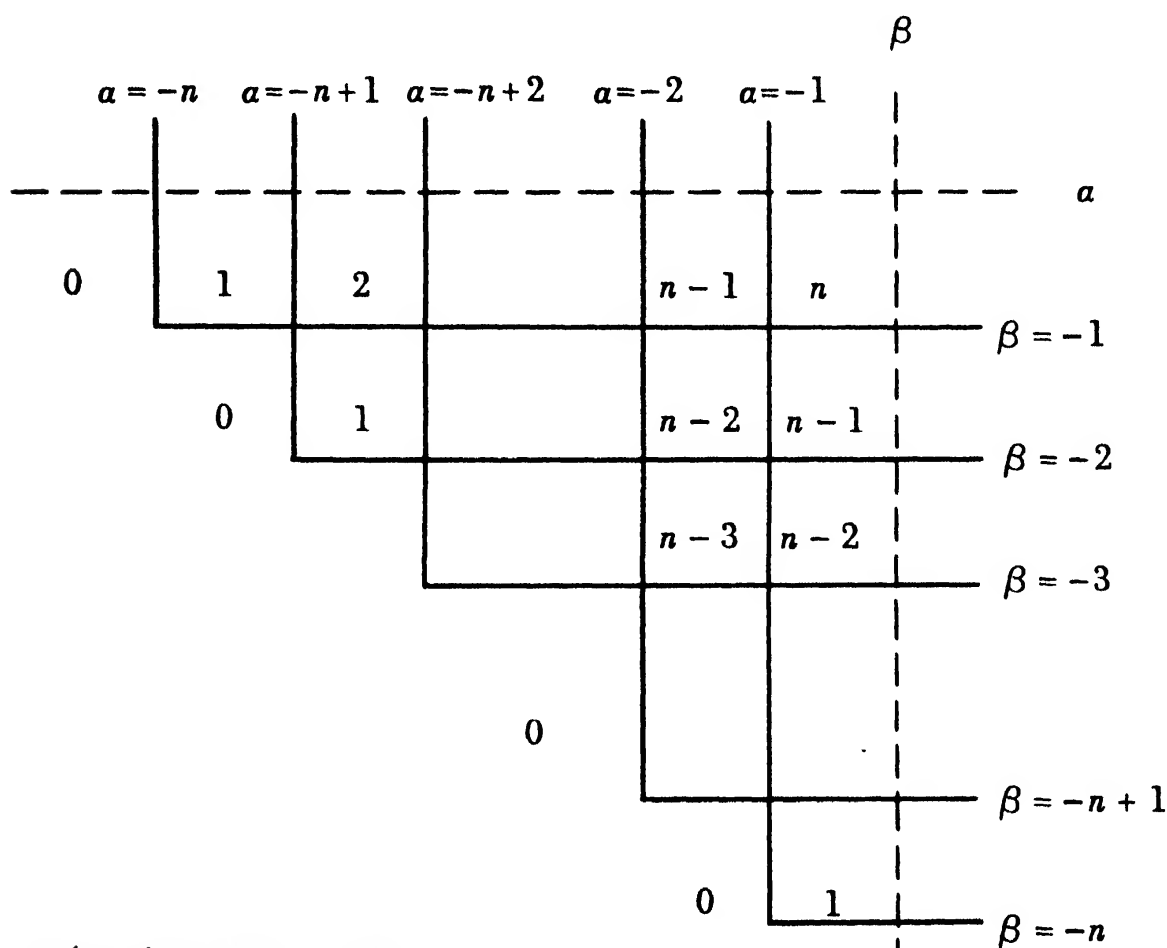
$$(21) \quad \mathcal{C}_0(z) = z^{-1} \sin z, \quad \mathcal{C}_{-1}(z) = z^{-2} \cos z$$

$$(22) \quad \mathcal{C}_{r+1}(z) = (2r+1) \mathcal{C}_r(z) - z^2 \mathcal{C}_{r-1}(z) \quad r = 0, 1, 2, \dots,$$

and the coefficients C_r also satisfy certain recurrence relations. The expansions (19) and (20) are convergent. They can also be used as asymptotic representations as $m \rightarrow \infty$, and for this purpose it is convenient to take $h = \frac{1}{2}$.

10.16. Zeros of Jacobi and related polynomials

Let us define Jacobi polynomials for all values of α, β, x by 10.8(12), and let us denote by $N_1(\alpha, \beta)$ the number of zeros of $P_n^{(\alpha, \beta)}(x)$ in the interval $(-1, 1)$. If $\alpha > -1$ and $\beta > -1$, Jacobi polynomials are orthogonal polynomials associated with the weight function 10.8(1), and by sec. 10.3 all their zeros are simple and located in $(-1, 1)$. For other real values of α and β the number of zeros in $(-1, 1)$ is indicated in the figure



$N_1(\alpha, \beta)$ for real α and β .

We see from 10.8(12) that for negative integer α , $P_n^{(\alpha, \beta)}(x)$ has a zero of order $|\alpha|$ at $x = 1$, and for negative integer β it has a zero of order $|\beta|$ at $x = -1$. In the interval $(-\infty, -1)$ there are $N_1(1-\alpha-\beta-2n, \beta)$ zeros, in the interval $(1, \infty)$ there are $N_1(1-\alpha-\beta-2n, \alpha)$ zeros. All zeros not accounted for in this enumeration occur in conjugate complex pairs.

Gegenbauer polynomials are defined by 10.9(18) for all values of λ, x . They are orthogonal polynomials, and all their zeros are simple and in the interval $(-1, 1)$, if $\lambda > -\frac{1}{2}$. For other real values of λ , the number of their zeros can be deduced from the result on Jacobi polynomials by means of 10.9(4).

The location of zeros of orthogonal Jacobi polynomials, and of their particular cases, in $(-1, 1)$ has been investigated by many authors. We refer the reader to Szegő (1939, Chapter VI), and for more recent work in particular to papers by Gatteschi; Geronimus; Lowan, Davids, and Levenson; and Tricomi listed at the end of this chapter.

We assume

$$(1) \quad \alpha > -1, \quad \beta > -1, \quad \lambda > -\frac{1}{2}, \quad x = \cos \theta \quad 0 < \theta < \pi,$$

and arrange the zeros in a monotonic sequence,

$$(2) \quad P_n^{(\alpha, \beta)}(\cos \theta_m) = 0, \quad 0 < \theta_1 < \theta_2 < \dots < \theta_n < \pi.$$

$$(3) \quad P_n^{(\alpha, \beta)}(x_m) = 0, \quad -1 < x_n < x_{n-1} < \dots < x_1 < 1, \quad x_m = \cos \theta_m.$$

For ultraspherical polynomials

$$(4) \quad x_m + x_{n-m} = 0$$

and hence it is sufficient to investigate the positive zeros ($1 \leq m \leq \frac{1}{2}n$). For Jacobi polynomials

$$x_m = x_m(\alpha, \beta, n)$$

and for Gegenbauer polynomials

$$x_m = x_m(\lambda, n) = x_m(\lambda - \frac{1}{2}, \lambda - \frac{1}{2}, n).$$

If m and n (and in the case of Jacobi polynomials also one of the parameters α, β) are fixed, we have the following monotonic properties:

$$(5) \quad x_m(\alpha, \beta, n) \downarrow -1 \quad \text{as} \quad \alpha \rightarrow \infty, \quad \uparrow 1 \quad \text{as} \quad \beta \rightarrow \infty$$

$m = 1, \dots, n$

$$(6) \quad x_m(\lambda, n) \downarrow 0 \quad \text{as} \quad \lambda \rightarrow \infty$$

$m = 1, \dots, [\frac{1}{2}n]$

The last of these relations, for instance, means that the m -th (positive) zero of a Gegenbauer polynomial is a strictly decreasing function of λ (for $\lambda > -\frac{1}{2}$) and tends to 0 as $\lambda \rightarrow \infty$. From (5) and (6) there follow corresponding statements for θ_n . Since we know $\theta_n(\pm \frac{1}{2}, \pm \frac{1}{2}, n)$ from 10.11(5) and 10.11(6), we have the following inequalities

$$(7) \quad (2m-1)\pi \leq (2n+1)\theta_n(a, \beta, n) \leq 2m\pi$$

$$-\frac{1}{2} \leq a, \quad \beta \leq \frac{1}{2}, \quad 1 \leq m \leq n$$

$$(8) \quad (m - \frac{1}{2})\pi/n < \theta_n(\lambda, n) < m\pi/(n+1) \quad 0 \leq \lambda \leq 1, \quad 1 \leq m \leq \frac{1}{2}n,$$

For further results see Szegő (1939, Chapter VI). Tricomi (1947) pointed out that the asymptotic behavior of the zeros of any function can be deduced from the asymptotic behavior of the function itself and has applied this principle to many functions, among them orthogonal polynomials (see Tricomi 1950, Gatteschi 1949, 1949a). It transpires that the asymptotic distribution of the zeros towards the middle of the interval depends on the zeros of trigonometric functions [see 1.14(5)] and the zeros near the end-points depend on the zeros of Bessel functions [see the remark following 10.14(12)].

Asymptotic formulas for the Christoffel numbers may be derived from asymptotic formulas for the zeros by means of 10.7(7).

For numerical values of the zeros and Christoffel numbers of Legendre polynomials see Lowan, Davids, and Levenson (1942, 1943).

10.17. Zeros of Laguerre and Hermite polynomials

The polynomial defined by 10.12(7) for all values of a and x , has n positive zeros if $a > -1$, $[n+a]$ positive zeros if $-n < a \leq -1$, no positive zeros if $a \leq -n$; it has a zero of order k at $x = 0$ when $a = -k$, $k = 1, 2, \dots, n$; and it has one negative zero if $(a+1)_n < 0$. All the zeros not accounted for in this enumeration occur in conjugate complex pairs. The Hermite polynomial of degree n has n real zeros which are situated symmetrically around the origin.

For detailed information on the location of the zeros of orthogonal Laguerre polynomials (i.e. for $a > -1$) and of Hermite polynomials we refer to Szegő (1939, Chapter VI), and to papers by Greenwood and Miller (1948), W. Hahn (1934), Salzer and Zucker (1949), Spencer (1937), and Tricomi.

We assume

$$(1) \quad a > -1, \quad > 0$$

and arrange the zeros of $L_n^\alpha(x)$ in increasing order so that

$$(2) \quad L_n^\alpha(x_m) = 0, \quad 0 < x_1 < x_2 < \cdots < x_n, \quad x_m = x_m(\alpha, n).$$

For fixed m, n we find again that x_m is an increasing function of α . For bounds for the zeros see Szegő (1939, Chapter VI), and W. Hahn (1934). The asymptotic representations of Laguerre and Hermite polynomials can be used to find approximations for the zeros (Tricomi 1949). It is clear from sec. 10.15 that we have to distinguish three cases. The "first" zeros are those for which m remains bounded while $n \rightarrow \infty$: these are investigated by means of 10.15(2). The "middle" zeros are those for which $|m - \frac{1}{2}n|$ remains bounded while $n \rightarrow \infty$: these are deduced from 10.15(8). The "last" zeros, for which $n - m$ remains bounded as $n \rightarrow \infty$, are deduced from 10.15(10). The resulting approximations give satisfactory numerical results even for moderate values of n , for instance $n = 10$.

Asymptotic formulas for Christoffel numbers may be derived from 10.7(7).

For numerical values of the zeros and Christoffel numbers of Laguerre polynomials, $L_n(x)$, see Salzer and Zucker (1949).

10.18. Inequalities for the classical polynomials

For inequalities for general orthogonal polynomials and for their application to the classical polynomials, see Szegő (1939, Chapter VII).

In the notation of sec. 10.3, there is the following result for monotonic weight functions (Szegő 1939, Theorem 7.2). If $w(x)$ is non-decreasing [non-increasing] and $b[a]$ is finite, then $[w(x)]^{1/2} |p_n(x)|$ attains its maximum in (a, b) at $b[a]$.

Application of this to those of the classical orthogonal polynomials whose weight function is monotonic, at once leads to the inequalities

$$(1) \quad |P_n(x)| \leq 1 \quad -1 \leq x \leq 1$$

$$(2) \quad [(1-x)/2]^{1/2 a + 1/2} |P_n^{(\alpha, 0)}(x)| \leq 1 \quad -1 \leq x \leq 1, \quad \alpha \geq -1/2$$

$$(3) \quad e^{-1/2 x} |L_n(x)| \leq 1 \quad x \geq 0.$$

Another fruitful source of inequalities is the Sonine-Pólya theorem (Szegő 1939, Theorem 7.31.1 and footnote). If in the differential equation

$$(4) \quad [k(x) y']' + \phi(x) y = 0$$

$k(x)$ and $\phi(x)$ are positive and continuously differentiable, and if $k(x) \phi(x)$ is monotonic, then the successive (relative) maxima of $|y|$ form an increasing or decreasing sequence according as $k(x) \phi(x)$ is decreasing or increasing.

The following results can be proved by constructing the differential equation satisfied by the functions involved, and applying the Sonine-Pólya theorem.

The successive maxima of $|P_n(x)|$, $n \geq 2$, as x increases from 0 to 1 form an increasing sequence. [This confirms (1).] The successive maxima of $(\sin \theta)^{1/2} |P_n(\cos \theta)|$, $n \geq 2$, as θ increases from 0 to $\frac{1}{2}\pi$ form an increasing sequence. As an application, it can be proved that

$$(5) \quad (\sin \theta)^{1/2} |P_n(\cos \theta)| < (1/2 \pi n)^{-1} \quad 0 \leq \theta \leq \pi.$$

Furthermore,

$$(6) \quad |P'_n(x)| \leq 1/2 n(n+1) \quad -1 \leq x \leq 1.$$

For Gegenbauer polynomials

$$(7) \quad \max_{-1 \leq x \leq 1} |C_n^\lambda(x)| = C_n^\lambda(1) = \frac{(2\lambda)_n}{n!} \quad \lambda > 0$$

$$(8) \quad \max_{-1 \leq x \leq 1} |C_{2m}^\lambda(x)| = |C_{2m}^\lambda(0)| = \frac{(\lambda)_m}{m!} \quad -m < \lambda < 0, \quad \lambda \text{ not integer}$$

$$(9) \quad \max_{-1 \leq x \leq 1} |C_{2m+1}^\lambda(x)| < 2[(2m+1)(2\lambda+2m+1)]^{-1/2} |(\lambda)_{m+1}|/m! \\ -m - 1/2 < \lambda < 0, \quad \lambda \text{ not integer.}$$

$$(10) \quad (\sin \theta)^\lambda |C_n^\lambda(\cos \theta)| < (1/2 n)^{\lambda-1} [\Gamma(\lambda)]^{-1} \quad 0 < \lambda < 1, \quad 0 \leq \theta \leq \pi.$$

For Jacobi polynomials we put

$$(11) \quad q = \max(a, \beta)$$

and obtain

$$(12) \quad \max_{-1 \leq x \leq 1} |P_n^{(\alpha, \beta)}(x)| = \max P_n^{(\alpha, \beta)}(\pm 1) = \binom{n+q-1}{n} \\ a > -1, \quad \beta > -1, \quad q \geq -1/2.$$

If $-1 < \alpha, \beta < -1/2$, the largest maximum of $|P_n^{(\alpha, \beta)}(x)|$ is one of the two nearest to $x_0 = (\beta - \alpha)/(\alpha + \beta + 1)$, and this maximum is of the order of $n^{-1/2}$ as $n \rightarrow \infty$. Among the numerous estimates for large n we mention only

$$(13) \quad \frac{d^m}{dx^m} P_n^{(\alpha, \beta)}(x) = O(n^q), \quad q = \max(2m + \alpha, 2m + \beta, m - 1/2) \\ n \rightarrow \infty.$$

For the special Laguerre polynomial L_n^0 we already have (3). Bounds for L_n^a may be obtained from this by using 10.12(39) with $\beta = 0$. The result is

$$(14) \quad |L_n^a(x)| \leq (a+1)_n (n!)^{-1} e^{\frac{1}{2}x} \quad a \geq 0$$

$$(15) \quad |L_n^a(x)| \leq [2 - (a+1)_n (n!)^{-1}] e^{\frac{1}{2}x} \quad -1 < a < 0.$$

The following results can be proved by applying the Sonine-Pólya theorem to the differential equation satisfied by the functions involved.

For any real a , the successive maxima of

$$e^{-\frac{1}{2}x} x^{\frac{1}{2}a+\frac{1}{2}} |L_n^a(x)|$$

form an increasing sequence provided that $2n + a + 1 > 1$ and

$$x > \max\{0, (a^2 - 1)/(2n + a + 1)\}.$$

The successive maxima of

$$e^{-\frac{1}{2}x} x^{\frac{1}{2}a+\frac{1}{2}} |L_n^a(x)|$$

form an increasing sequence provided that $x > 0$ and

$$x^2 > \max(0, a^2 - \frac{1}{4}).$$

The successive maxima of

$$e^{-\frac{1}{2}x} |L_n^a(x)|$$

form a decreasing sequence when $a > -1$,

$$0 \leq x < (2a+1)(2n+a+1)/(a+1)$$

and an increasing sequence when $a > -1$,

$$x > (2a+1)(2n+a+1)/(a+1).$$

The successive maxima of

$$e^{-\frac{1}{2}x} x^{\frac{1}{2}a} L_n^a(x)$$

form a decreasing sequence if

$$0 < x < 2n + a + 1,$$

and an increasing sequence when

$$x > 2n + a + 1 > 0.$$

All these statements are contained in the following more general result. For real a and β , the successive maxima for $x > 0$ of

$$e^{-\frac{1}{2}x} x^\beta |L_n^a(x)|$$

form an increasing or decreasing sequence according as

$$4\beta(\beta - \alpha)(\alpha - 2\beta) + (2n + \alpha + 1)(2\alpha - 4\beta + 1)x - (\alpha - 2\beta + 1)x^2$$

is negative or positive.

For an asymptotic estimate see Szegő (1939, Theorem 7.6.4); improvements of this estimate may be derived from Tricomi's expansion 10.15(4).

Bounds for Hermite polynomials may be derived from (14) and (15) by means of 10.13(2) and 10.13(4). See also Sansone (1950a).

$$(16) \quad \exp(-\tfrac{1}{2}x^2) |H_{2m}(x)| \leq 2^{2m} m! (2 - g_m)$$

$$(17) \quad x^{-1} \exp(-\tfrac{1}{2}x^2) |H_{2m+1}(x)| \leq 2^{2m+2} (m+1)! g_{m+1}$$

where

$$(18) \quad g_n = (\tfrac{1}{2})_n / n! = (\pi n)^{-1/2} + O(n^{-3/2}).$$

H. Cramér has proved

$$(19) \quad \exp(-\tfrac{1}{2}x^2) |H_n(x)| < k 2^{1/2n} (n!)^{1/2}$$

where k is a constant for which Charlier (1931) gave the approximation 1.086435. Sansone (1950) gave bounds valid for complex values of the variable.

From the Sonine-Pólya theorem it may be proved that the successive maxima of $|H_n(x)|$, and likewise those of $\exp(-\tfrac{1}{2}x^2) |H_n(x)|$, for $x \geq 0$ form an increasing sequence.

Let $\mu_{r,n}$ be the r -th (relative) maximum of $f(x) |p_n(x)|$, where $f(x)$ is a fixed non-negative function and $\{p_n(x)\}$ is a sequence of orthogonal polynomials. The results derived from the Sonine-Pólya theorem state monotonic properties of $\mu_{r,n}$ as r increases while n is fixed. The study of numerical tables led John Todd to some conjectures about monotonic properties of $\mu_{r,n}$ for fixed r and increasing n . The following results were subsequently proved. For

$$f(x) = 1, \quad p_n(x) = P_n(x),$$

and counting maxima from $x = 1$ (to the left), Cooper (1950) proved that $\mu_{r,n}$ is a decreasing function of n for sufficiently large n , and Szegő (1950) proved that this true for all $n \geq r + 1$. For

$$f(x) = 1, \quad p_n(x) = C_n^\lambda(x),$$

Szász (1950) proved that $n! \mu_{r,n} / \Gamma(n + 2\lambda)$ is a decreasing function of n . For

$$f(x) = e^{-1/2x^2} \quad p_n(x) = L_n(x).$$

J. Todd (1950) proved that $\mu_{r,n}$ is an increasing or decreasing function of n as r is odd or even.

P. Turán observed that

$$u_n = P_n(x) \quad -1 \leq x \leq 1$$

satisfies the inequality

$$(20) \quad u_n^2 - u_{n-1} u_{n+1} \geq 0.$$

Szegő (1948) gave several proofs of this inequality and showed that it is also satisfied by

$$u_n = C_n^\lambda(x)/C_n^\lambda(1) = n! C_n^\lambda(x)/(2\lambda)_n \quad -1 \leq x \leq 1$$

$$u_n = L_n^\alpha(x)/L_n^\alpha(0) = n! L_n^\alpha(x)/(a+1)_n \quad > 0$$

$$u_n = H_n(x).$$

These results have been reproved, refined, and generalized; determinants whose elements are orthogonal polynomials have been considered, and other related investigations have been carried out by Madhava Rao and Thiruvengkatachar (1949), Sansone (1949), Szász (1950a, 1951), Beckenbach, Seidel, and Szász (1951), Forsythe (1951). See also J. L. Burchall (1951, 1952).

10.19. Expansion problems

The expansion of a given, "arbitrary" or analytic function in a series of orthogonal polynomials has been discussed often and in great detail. The subject is somewhat outside the scope of the present survey, and a brief indication of some of the more important results must suffice. For further information see Szegő (1939, especially Chapter IX), Kaczmarz and Steinhaus (1935).

Let $\{p_n(x)\}$ be a system of orthogonal polynomials belonging to the weight function $w(x)$ on the interval (a, b) . We assume that the assumptions of sections 10.1 and 10.2 are satisfied, and denote by L^p , $p \geq 1$ the class of functions $f(x)$ for which the (Lebesgue) integral

$$\int_a^b |f(x)|^p w(x) dx$$

exists and is finite. We put

$$(1) \quad h_n = \int_a^b [p_n(x)]^2 w(x) dx$$

and call

$$(2) \quad a_n = h_n^{-1} \int_a^b f(x) p_n(x) dx$$

the Fourier coefficients,

$$(3) \quad \sum a_n p_n(x)$$

the (generalized) Fourier series of $f(x)$ with respect to the system $\{p_n(x)\}$ of orthogonal polynomials. We shall say that the series (3) *converges in L^2_w* to $f(x)$ if

$$(4) \quad \int_a^b |f(x) - s_n(x)|^p w(x) dx \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

where $s_n(x)$ is the n -th partial sum of (3).

Approximation in L^2_w has already been discussed in sec. 10.2, and from the results described there it follows that in case of a *finite* interval (a, b) for any function $f(x)$ of L^2_w , (3) converges in L^2_w to $f(x)$. Convergence in L^p_w has been investigated by Pollard (1946, 1947, 1948, 1949) and Wing (1950). For Jacobi polynomials, given by 10.8(1), Pollard proved convergence in L^p_w when

$$(5) \quad \alpha \geq -\frac{1}{2}, \quad \beta \geq -\frac{1}{2}$$

and

$$(6) \quad 4 \max \left(\frac{\alpha+1}{2\alpha+3}, \frac{\beta+1}{2\beta+3} \right) < p < 4 \min \left(\frac{\alpha+1}{2\alpha+1}, \frac{\beta+1}{2\beta+1} \right).$$

For Gegenbauer polynomials we have 10.9(1) and convergence in L^p_w when

$$(7) \quad \lambda > 0, \quad \frac{2\lambda+1}{\lambda+1} < p < \frac{2\lambda+1}{\lambda}.$$

Lastly, for Legendre polynomials $w(x) = 1$ and we have convergence in L^p_w when

$$(8) \quad \frac{4}{3} < p < 4.$$

It has been pointed out in sec. 10.2 that infinite intervals present additional difficulties. Nevertheless, convergence in L^p_w for Laguerre polynomials with $\alpha > -1$, and for Hermite polynomials, has been proved when $p = 2$.

The series (3) is said to converge to $f(x)$ for a fixed x , or in an interval, if for that x , or for all x in that interval

$$s_n(x) \rightarrow f(x) \quad \text{as} \quad n \rightarrow \infty,$$

where $s_n(x)$ is again the n -th partial sum of (3). This type of convergence (sometimes called "point-wise convergence") requires much more restrictive assumptions on $f(x)$ than convergence in L^p_μ .

Rau (1950) has investigated the convergence of the expansion of a function $f(x)$ in a series of Jacobi polynomials with $\alpha > -1$, $\beta > -1$. Assuming that $f(x)$ is continuous and has a piece-wise continuous derivative, he proved that the expansion converges to $f(x)$ uniformly in every interval $-1 + \epsilon \leq x \leq 1 - \epsilon$, $\epsilon > 0$.

The Abel summability of series of Laguerre polynomials was investigated by Caton and Hille (1945) by means of Laplace integrals.

Asymptotic formulas such as 10.14(1), (7), (10), and 10.15(1), (18) suggest a connection between the convergence of orthogonal expansions and that of certain related Fourier series. This is the source of the so-called equiconvergence theorems. As a sample, we shall give an equiconvergence theorem for Legendre polynomials (Haar, 1918).

Let $|f(x)|^2$ be integrable in $(-1, 1)$; let $s_n(x)$ be the n -th partial sum of the expansion of $f(x)$ in Legendre polynomials, and let $\sigma_n(\theta)$ be the n -th partial sum of the Fourier cosine expansion of $f(\cos \theta)$. Then

$$s_n(\cos \theta) - \sigma_n(\theta) \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty, \quad 0 < \theta < \pi.$$

Such equiconvergence theorems, in combination with conditions for the convergence of Fourier series, enable one to discuss the convergence of orthogonal expansions. Equiconvergence theorems for Jacobi, Laguerre, and Hermite polynomials were given by Szegő (1939, Chapter IX). Szegő also gives some results regarding the behavior of such series at the end-points of the basic interval.

The expansion of analytic functions presents rather different problems. A series of Jacobi polynomials converges in an ellipse whose foci are at ± 1 , and every function which is analytic in such an ellipse may be expanded in a series of Jacobi polynomials ($\alpha, \beta > -1$) there. A function which is analytic outside such an ellipse, and vanishes at infinity, may be expanded in a series of Jacobi functions of the second kind, $Q_n^{(\alpha, \beta)}$, there ($\alpha, \beta > -1$, see Szegő sec. 9.2).

In the case of Laguerre polynomials the region of convergence is a parabola around the positive real axis, with its focus at the origin: in the case of Hermite polynomials the region of convergence is a strip whose central line is the real axis. In both cases the region of convergence is unbounded and an analytic function which is to be expanded in a series of Laguerre or Hermite polynomials must satisfy certain growth conditions in addition to being analytic in an appropriate region. Expansions in series of Laguerre polynomials were investigated by Pollard

(1947a), series of Hermite polynomials by Giuliotto (1939), and Hille (1939, 1939a, 1940).

10.20. Examples of expansions

In this section we list some series of orthogonal polynomials whose sum can be given in closed form. Except in the case of Legendre, Hermite, and Laguerre polynomials, not many such series are known, and some of the following examples have been developed by Tricomi to fill this gap. The computation of the coefficients of such an expansion is based on 10.19(2), where one may often take advantage of Rodrigues' formula (or its generalizations) to simplify the integral by integration by parts in the manner explained in the second paragraph of sec. 10.7.

In the following formulas we shall freely use the notations for confluent hypergeometric and related functions which have been introduced in Chapters VI, VIII, IX.

SERIES OF JACOBI POLYNOMIALS

Notations as in sec. 10.8. We always assume $\alpha, \beta > -1$, and use h_n as defined in 10.8(4).

$$(1) \quad \operatorname{sgn} x = c_0 + \sum_{n=1}^{\infty} \frac{1}{nh_n} P_{n-1}^{(\alpha+1, \beta+1)}(0) P_n^{(\alpha, \beta)}(x) \quad -1 < x < 1.$$

Here

$$(2) \quad \operatorname{sgn} x = \begin{cases} 1 & \text{when } x > 0, \\ -1 & \text{when } x < 0, \end{cases}$$

and

$$c_0 = \frac{\Gamma(\alpha + \beta + 2)}{\Gamma(\alpha + 1)\Gamma(\beta + 1)} 2^{-\alpha-\beta-1} \int_0^1 [(1-x)^\alpha (1+x)^\beta - (1+x)^\alpha (1-x)^\beta] dx.$$

Note that only the terms corresponding to odd values of n actually occur in the summation in (1).

$$(3) \quad (1-x)^\rho = 2^\rho \Gamma(\alpha + \rho + 1) \times \sum_{n=0}^{\infty} \frac{\Gamma(2n + \alpha + \beta + 1) \Gamma(n + \alpha + \beta + 1) (-\rho)_n}{\Gamma(n + \alpha + 1) \Gamma(n + \alpha + \beta + \rho + 2)} P_n^{(\alpha, \beta)}(x) \\ -\rho < \min(\alpha + 1, \tfrac{1}{2}\alpha + \tfrac{3}{4}), \quad -1 < x < 1$$

$$(4) \quad e^{ixy} = (2iy)^{-\frac{1}{2}(\alpha+\beta)-1} \sum_{n=0}^{\infty} \frac{\Gamma(n+\alpha+\beta+1)}{\Gamma(2n+\alpha+\beta+1)} M_{k,n}(2iy) P_n^{(\alpha,\beta)}(x)$$

$$-1 < x < 1$$

where

$$k = \frac{1}{2}(\alpha - \beta), \quad m = n + \frac{1}{2}(\alpha + \beta + 1).$$

For a generating function see 10.8(29); for a bilinear generating function see Watson (1934), Erdélyi (1937a) and Bailey; for another expansion in products of Jacobi polynomials see Bateman (1904, 1905).

SERIES OF GEGENBAUER POLYNOMIALS

Notation as in sec. 10.9. The constant h_n is defined by 10.9(7).

$$(5) \quad \operatorname{sgn} x = 4 \sum_{n=1}^{\infty} \frac{(-1)^n (\lambda)_{n+1}}{(2n+1)(2n+2\lambda+1) n! h_{2n+1}} C_{2n+1}^{\lambda}(x)$$

$$\lambda > -\frac{1}{2}, \quad -1 < x < 1.$$

$$(6) \quad (1-x)^{\rho} = 2^{2\lambda+\rho} \pi^{-\frac{1}{2}} \Gamma(\lambda) \Gamma(\lambda+\rho+\frac{1}{2})$$

$$\times \sum_{n=0}^{\infty} \frac{(n+\lambda) (-\rho)_n}{\Gamma(n+2\lambda+\rho+1)} C_n^{\lambda}(x)$$

$$-1 < x < 1, \quad -\rho < \frac{1}{2}(\lambda+1) \text{ if } \lambda \geq 0, \quad -\rho < \frac{1}{2} + \lambda \text{ if } -\frac{1}{2} < \lambda \leq 0.$$

$$(7) \quad e^{ixy} = \Gamma(\lambda) (\frac{1}{2}y)^{-\lambda} \sum_{n=0}^{\infty} i^n (n+\lambda) J_{n+\lambda}(y) C_n^{\lambda}(x)$$

$$-1 < x < 1, \quad \lambda > 0$$

$$(8) \quad (y \sin \phi \sin \theta)^{\frac{1}{2}-\lambda} J_{\lambda-\frac{1}{2}}(y \sin \phi \sin \theta) e^{iy \cos \phi \cos \theta}$$

$$= 2^{\frac{1}{2}} y^{-\lambda} \Gamma(\lambda) \sum_{n=0}^{\infty} i^n \frac{n! (n+\lambda)}{(2\lambda)_n \Gamma(2n+\lambda)}$$

$$\times J_{n+\lambda}(y) C_n^{\lambda}(\cos \phi) C_n^{\lambda}(\cos \theta)$$

$$0 < \phi, \theta < \pi, \quad \lambda > 0.$$

$$(9) \quad \omega^{-\lambda} C_{\lambda}(\omega) = 2^{\lambda} \Gamma(\lambda) \sum_{n=0}^{\infty} (n+\lambda) z^{-\lambda} Z^{-\lambda}$$

$$\times J_{n+\lambda}(z) C_{n+\lambda}(Z) C_n^{\lambda}(\cos \phi)$$

where

$$|ze^{\pm i\phi}| < |Z|, \quad \omega^2 = z^2 + Z^2 - 2zZ \cos \phi,$$

and

$$C_{\lambda}(\omega) = c_1 J_{\lambda}(\omega) + c_2 J_{-\lambda}(\omega)$$

is any cylinder function in the sense of Sonine and Watson (Watson, 1922, sec. 3.9). In the case $c_2 = 0$ the restrictions on z, Z may be omitted.

Some expansions in series of Gegenbauer polynomials have been noted in sec. 10.9: for a bilinear generating function see Watson (1933b).

SERIES OF LEGENDRE POLYNOMIALS

Notations as in sec. 10.10. The constant g_n is defined in 10.10(4). All expansions valid for $-1 < x < 1$, or $0 < \theta < \pi$, respectively, unless stated differently.

$$(10) \quad |x|^{\rho} = \sum_{n=0}^{\infty} (-1)^n \frac{(2n + \frac{1}{2})(-\frac{1}{2}\rho)_n}{(\frac{1}{2}\rho + \frac{1}{2})_{n+1}} P_{2n}(x) \quad \rho > -1$$

$$(11) \quad |x|^{\rho} \operatorname{sgn} x = \sum_{n=0}^{\infty} (-1)^n (2n + \frac{3}{4}) \frac{(\frac{1}{2} - \frac{1}{2}\rho)_n}{(1 + \frac{1}{2}\rho)_{n+1}} P_{2n+1}(x) \quad \rho > -1$$

$$(12) \quad (1-x)^{\rho} = 2^{\rho} \sum_{n=0}^{\infty} \frac{2n+1}{n+\rho+1} \frac{(-\rho)_n}{(1+\rho)_n} P_n(x) \quad \rho > -\frac{3}{4}$$

$$(13) \quad (1-x^2)^{\frac{1}{2}} = \frac{1}{2}\pi \left[\frac{1}{2} - \sum_{n=1}^{\infty} \frac{4n+1}{(2n-1)(2n+2)} g_n^2 P_{2n}(x) \right]$$

$$(14) \quad \frac{1}{2} e^{-\frac{1}{2}\phi} [\cos^2(\frac{1}{2}\phi) - \cos^2(\frac{1}{2}\theta)]^{-\frac{1}{2}} = \sum_{n=0}^{\infty} e^{in\phi} P_n(\cos \theta) \quad 0 \leq \phi < \theta < \pi$$

$$(15) \quad \log[1 + \csc(\frac{1}{2}\theta)] = \sum_{n=0}^{\infty} (n+1)^{-1} P_n(\cos \theta).$$

Series involving Bessel functions may be derived from (7), (8), and (9) by putting $\lambda = \frac{1}{2}$. Generating functions are listed in 10.10(v) and (viii).

SERIES OF LAGUERRE POLYNOMIALS

Notations as in sec. 10.12. We always assume $a > -1, x > 0$.

$$(16) \quad x^{\rho} = \Gamma(a+\rho+1) \sum_{n=0}^{\infty} \frac{(-\rho)_n}{\Gamma(a+n+1)} L_n^a(x) \quad -\rho < 1 + \min(a, \frac{1}{2}a - \frac{1}{4})$$

$$(17) \quad \psi(a+1) - \log x = \Gamma(a+1) \sum_{n=1}^{\infty} \frac{(n-1)!}{\Gamma(a+n+1)} L_n^a(x)$$

$$(18) \quad -e^{x+y} \text{Ei}[-\max(x, y)] = \sum_{n=0}^{\infty} (n+1)^{-1} L_n(x) L_n(y) \quad x, y > 0$$

$$(19) \quad e^x x^{-a} \Gamma(a, x) = \sum_{n=0}^{\infty} (n+1)^{-1} L_n^a(x)$$

$$(20) \quad e^{x+y} (xy)^{-a} \Gamma[a, \max(x, y)] \gamma[a, \min(x, y)] \\ = \sum_{n=0}^{\infty} \frac{n!}{(n+1)(a)_{n+1}} L_n^a(x) L_n^a(y)$$

$$(21) \quad (xy)^{-a} e^{x+y} \{ \Gamma[a, \max(x, y)] - \Gamma(a, x) \Gamma(a, y) / \Gamma(a) \} \\ = \sum_{n=0}^{\infty} \frac{n!}{(n+1) \Gamma(n+a+1)} L_n^a(x) L_n^a(y) \quad x, y > 0$$

$$(22) \quad e^{\min(x, y)} = 1 + \sum_{n=1}^{\infty} [L_n(x) - L_{n-1}(x)] [L_n(y) - L_{n-1}(y)]$$

$$(23) \quad (xy)^{-\frac{1}{2}a} e^{-\frac{1}{2}(x+y)} e^{-a\pi i} \gamma[a, e^{i\pi} \min(x, y)] \\ = \sum_{n=0}^{\infty} \frac{n!}{(n+a) \Gamma(n+a+1)} L_n^a(x) L_n^a(y) \quad \text{Re } a > 0$$

$$(24) \quad \frac{\Gamma(a+1, x)}{\Gamma(a+1)} - H(x-y) = y^{a+1} e^{-y} \sum_{n=1}^{\infty} \frac{(n-1)!}{\Gamma(a+n+1)} L_{n-1}^{a+1}(y) L_n^a(x) \\ 0 < x, y$$

In (24), $H(z) = 0, \frac{1}{2}, 1$ according as $z < 0, = 0, > 0$.

$$(25) \quad x^{\frac{1}{2}(a-\beta)} y^{-\frac{1}{2}(a+\beta)} e^y J_{a+\beta}[2(xy)^{\frac{1}{2}}] = \sum_{n=0}^{\infty} \frac{(-1)^n n!}{\Gamma(a+n+1)} L_n^a(x) L_n^{\beta-n}(y)$$

$$(26) \quad \Gamma(a) \Psi(a, a+1; x) = \sum_{n=0}^{\infty} (n+a)^{-1} L_n^a(x) \quad a < \frac{1}{2}$$

$$(27) \quad (1-y)^{-a} \Phi\left(a, c; \frac{xy}{y-1}\right) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n} y^n L_n^{c-1}(x) \\ x > 0, \quad |y| < 1, \quad c > 0.$$

Other series of Laguerre polynomials in 10.12(v) and (vii). The expansion 10.14(11) is an expansion in Laguerre polynomials when β is an integer $\geq -n$.

SERIES OF HERMITE POLYNOMIALS

Notations as in sec. 10.13.

$$(28) \quad |x|^\rho = \frac{\Gamma(\frac{1}{2} + \frac{1}{2}\rho)}{\pi^{\frac{1}{2}}} \sum_{n=0}^{\infty} (-1)^n \frac{(-\frac{1}{2}\rho)_n}{(2n)!} H_{2n}(x) \quad \rho > -1$$

$$(29) \quad |x|^\rho \operatorname{sgn} x = \frac{\Gamma(1 + \frac{1}{2}\rho)}{\pi^{\frac{1}{2}}} \sum_{n=0}^{\infty} (-1)^n \frac{(\frac{1}{2} - \frac{1}{2}\rho)_n}{(2n+1)!} H_{2n+1}(x) \quad \rho > -1$$

$$(30) \quad \pi^{\frac{1}{2}} \operatorname{Erfi}[\min(x, y)] = \sum_{n=0}^{\infty} \frac{H_{2n+1}(x) H_{2n+1}(y)}{2^{2n+1} (2n+1)(2n+1)!} \quad x, y \geq 0$$

$$(31) \quad \exp(\frac{1}{4}x^2) D_{2\nu}(x) = \frac{2^\nu}{\Gamma(-\nu)} \sum_{n=0}^{\infty} \frac{(-1)^n H_{2n}(2^{-\frac{1}{2}}x)}{(n-\nu) 2^{2n} n!}$$

$$(32) \quad \exp(\frac{1}{4}x^2) D_{2\nu+1}(x) = \frac{2^{\nu+\frac{1}{2}}}{\Gamma(-\nu)} \sum_{n=0}^{\infty} \frac{(-1)^n H_{2n+1}(2^{-\frac{1}{2}}x)}{(n-\nu) 2^{2n+1} n!}$$

$$(33) \quad (1+y)^{-a} \Phi\left(a, \frac{1}{2}; \frac{x^2 y}{1+y}\right) = \sum_{n=0}^{\infty} \frac{(a)_n}{(2n)!} y^n H_n(x) \quad |y| < 1$$

$$(34) \quad 2x(1+y)^{-a} \Phi\left(a, \frac{3}{2}; \frac{x^2 y}{1+y}\right) = \sum_{n=0}^{\infty} \frac{(a)_n}{(2n+1)!} y^n H_{2n+1}(x).$$

Other series of Hermite polynomials are in 10.13(v).

The following key indicates the derivation of these examples; it also gives references to further material on infinite series of classical polynomials.

Series of Jacobi polynomials. The coefficients were computed by integrations by parts. For other examples see Brafman (1951).

(5) From (1) by 10.9(4).

(6) From (3) by 10.9(4).

(7) From (4) by 10.9(4); Watson (1922, p. 368).

(8) Watson (1922, p. 370).

(9) Watson (1922, p. 365).

Series of Legendre polynomials. Many examples may be obtained from series of Jacobi polynomials or series of Gegenbauer polynomials by means of 10.10(3). Numerous other examples are found in books on Legendre functions. For some examples, see Tricomi (1936, 1939-40).

- (16) Tricomi (1948, p. 332).
- (17) Toscano (1949).
- (18) Neumann (1912).
- (19) 9.4(5).
- (20) 9.4(4).
- (21) Watson, (1938).
- (22) Tricomi (1935), Doetsch (1935).
- (23) Erdélyi (1936).
- (24) Tricomi (1948).
- (25) Toscano (1949).
- (26) 6.12(3).
- (27) 6.12(5).

For some examples of series of Laguerre polynomials see Erdélyi (1937, 1938).

- (28), (29) From (16) by 10.13(2) and (3).
- (30) From (3) by 10.13(2) and (3).
- (31), (32) Tricomi (1950a).
- (33), (34) From (27) by 10.13(2) and (3).

10.21. Some classes of orthogonal polynomials

Beside the classical orthogonal polynomials there are other classes of special orthogonal polynomials which have been investigated in detail. In this section we shall describe some of these, mentioning very briefly those discussed in Szegő's book, and giving fuller details about those not otherwise conveniently accessible.

POLYNOMIALS OF S. BERNSTEIN AND G. SZEGŐ

These polynomials belong to the interval $(-1, 1)$ and their weight function $w(x)$ is of one of the forms

$$(1-x^2)^{-\frac{1}{2}} [\rho(x)]^{-1}, \quad (1-x^2)^{\frac{1}{2}} [\rho(x)]^{-1},$$

$$[(1-x)/(1+x)]^{\frac{1}{2}} [\rho(x)]^{-1}$$

where $\rho(x)$ is a polynomial of exact degree l , and positive for $-1 \leq x \leq 1$. Christoffel's formula 10.3(12) suggests a connection between these polynomials on the one side, and certain Jacobi polynomials on the other side.

The polynomials were encountered by Szegő (1921) and investigated by Bernstein (1930, 1932). See Szegő (1939) sec. 2.6.

POLYNOMIALS OF E. HEINE AND N. ACHYESER

Heine's polynomials belong to the interval $(0, a)$ and to the weight function

$$(1) \quad w(x) = [x(a-x)(b-x)]^{-\frac{1}{2}} \quad 0 < a < b.$$

They are related to Jacobian elliptic functions.

Heine (1878-1881, vol. 1, p. 294-296) showed that the polynomial of degree n satisfies a differential equation of the form

$$(2) \quad 2\psi(x)(x-\gamma)\frac{d^2\gamma}{dx^2} + [(x-\gamma)\psi'(x) - 2\psi(x)]\frac{d\gamma}{dx} \\ + [a + \beta x - n(2n-1)x^2]\gamma = 0$$

where

$$\psi(x) = x(a-x)(b-x)$$

and a, β, γ are certain constants. This differential equation has four singularities of the regular type and hence is an instance of Heun's equation.

Achyeser (1934) investigated the orthogonal polynomials associated with the interval $(-1, 1)$ and the weight function

$$w(x) = \begin{cases} |c-x|[(1-x^2)(a-x)(b-x)]^{-\frac{1}{2}} & -1 < x < a \text{ or } b < x < 1 \\ 0 & a < x < b. \end{cases}$$

Here $-1 < a < b < 1$, and c depends on a and b . These polynomials are also related to elliptic functions.

POLYNOMIALS OF F. POLLACZEK

Recently, F. Pollaczek defined certain families of orthogonal polynomials which are generalizations of classical orthogonal polynomials. The weight functions associated with Pollaczek's polynomials fail to satisfy certain conditions which it is customary to impose in the general theory (roughly speaking, they vanish too strongly at the end-points of the interval), and thus these polynomials are important, and readily accessible, examples of certain "irregular" phenomena in the general theory of orthogonal polynomials.

Finite interval. Let a, b, λ be real parameters, $a \geq |b|$, $\lambda > -1$. We put

$$(3) \quad -1 \leq x = \cos \theta \leq 1 \quad 0 \leq \theta \leq \pi,$$

and use the abbreviation

$$(4) \quad t = (a \cos \theta + b)/\sin \theta = (ax + b)(1-x^2)^{-\frac{1}{2}}.$$

The polynomials $P_n^\lambda(x; a, b)$ are defined recurrently.

$$(5) \quad P_{-1}^\lambda = 0, \quad P_0^\lambda = 1$$

$$(6) \quad n P_n^\lambda - 2[(n-1+\lambda+a)x + b] P_{n-1}^\lambda + (n+2\lambda-2) P_{n-2}^\lambda = 0$$

$$n = 1, 2, \dots$$

These polynomials were defined by Pollaczek (1949a, for $\lambda = \frac{1}{2}$, 1949c, for $\text{Re } \lambda > 0$) and studied by Szegő (1950a). Some related polynomials were also studied by Pollaczek (1949b, 1950a).

Multiplying (6) by z^n and adding, one obtains a simple differential equation of the first order for the generating function, and hence

$$(7) \quad \sum_{n=0}^{\infty} P_n^\lambda(x; a, b) z^n = (1 - ze^{i\theta})^{-\lambda+it} (1 - ze^{-i\theta})^{-\lambda-it} \quad |z| < 1.$$

Comparison with 10.9 (29) and 10.10 (39) shows the relation to Gegenbauer and Legendre polynomials

$$(8) \quad P_n^\lambda(x; 0, 0) = C_n^\lambda(x), \quad P_n^{\frac{1}{2}}(x; 0, 0) = P_n(x).$$

The polynomials are orthogonal on the interval (4), the weight function being

$$(9) \quad w^{(\lambda)}(x; a, b) = \pi^{-1} 2^{2\lambda-1} e^{(2\theta-\pi)t} (\sin \theta)^{2\lambda-1} |\Gamma(\lambda+it)|^2.$$

The asymptotic behavior of $P_n^{\frac{1}{2}}(x; a, b)$ when x is fixed, between -1 and 1 , and $n \rightarrow \infty$ was investigated by Szegő.

Either from the generating function (7), or from the recurrence relation (6) it may be proved that

$$(10) \quad n! P_n^\lambda(x; a, b) = (2\lambda)_n e^{in\theta} {}_2F_1(-n, \lambda+it; 2\lambda; 1 - e^{-2i\theta}).$$

This expression in terms of hypergeometric polynomials is the source of many further formulas for Pollaczek's polynomials. It should be noted that t depends on x so that P_n^λ does not satisfy any differential equation. Formulas connecting P_n^λ for different values of λ follow from (10) as instances of relations between contiguous hypergeometric series.

In a later paper (1950c), Pollaczek introduced a more general system of polynomials which depends on the real parameters a, b, c, λ where

$$(11) \quad \text{either } a > |b|, \quad 2\lambda + c > 0, \quad c \geq 0$$

$$\text{or } a > |b|, \quad 2\lambda + c \geq 1, \quad c > -1.$$

With the notations (3) and (4), $P_n^\lambda(x; a, b, c)$ satisfies

$$(12) \quad P_{-1}^\lambda = 0, \quad P_0^\lambda = 1$$

$$(13) \quad (n+c)P_n^\lambda - 2[(n-1+\lambda+a+c)x+b]P_{n-1}^\lambda + (n+2\lambda+c-2)P_{n-2}^\lambda = 0$$

$$n = 1, 2, \dots$$

The generating function of these polynomials has been obtained by Pollaczek, who also proved that these polynomials are orthogonal on the interval (4), the weight function being

$$(14) \quad w^{(\lambda)}(x; a, b, c) = \frac{(2 \sin \theta)^{2\lambda-1} e^{(2\theta-\pi)t}}{2\pi \Gamma(2\lambda+c) \Gamma(c+1)}$$

$$\times |\Gamma(\lambda+c+it)|^2 |{}_2F_1(1-\lambda+it, c; c+\lambda+it; e^{2i\theta})|^{-2}.$$

The recurrence relation (13) is a difference equation for P as a function of n . This equation serves to express $P_n^\lambda(x; a, b, c)$ in terms of hypergeometric functions. The expression is fairly complicated and the hypergeometric series appearing in it are no longer polynomials. Putting

$$A_n = \frac{\Gamma(2\lambda+c+n)}{\Gamma(c+n+1) \Gamma(2\lambda)} e^{i(c+n)\theta} {}_2F_1(-c-n, \lambda+it; 2\lambda; 1-e^{-2i\theta}),$$

$$B_n = \frac{\Gamma(1-\lambda+it) \Gamma(1-\lambda-it)}{\Gamma(2-2\lambda)} (2 \sin \theta)^{1-2\lambda} e^{i(2\lambda+c+n-1)\theta}$$

$$\times {}_2F_1(1-2\lambda-c-n, 1-\lambda+it; 2-2\lambda; 1-e^{-2i\theta})$$

the resulting expression is

$$(15) \quad P_n^\lambda(x; a, b, c) = \frac{A_{-1} B_n - A_n B_{-1}}{A_{-1} B_0 - A_0 B_{-1}}.$$

In this form, it is valid when 2λ is not an integer. An alternative form, valid for integer values of 2λ is available. $A_{-1} = 0$ when $c = 0$, and in this case (15) reduces to (10).

Infinite interval. For the infinite interval $-\infty < x < \infty$, Pollaczek (1950b) has the system of polynomials $P_n^\lambda(x; \varphi)$ where

$$(16) \quad \lambda > 0, \quad 0 < \varphi < \pi$$

are parameters,

$$(17) \quad P_{-1}^\lambda = 0, \quad P_0^\lambda = 1,$$

$$(18) \quad n P_n^\lambda - 2[(n-1+\lambda) \cos \varphi + x \sin \varphi] P_{n-1}^\lambda + (n-2+2\lambda) P_{n-2}^\lambda = 0$$

$$n = 1, 2, \dots$$

Clearly, these polynomials may be obtained from those defined by (6) by replacing θ by φ and t by x . The generating function is

$$(19) \quad \sum_{n=0}^{\infty} P_n^\lambda(x; \varphi) z^n = (1 - ze^{i\varphi})^{-\lambda+ix} (1 - ze^{-i\varphi})^{-\lambda-ix} \quad |z| < 1,$$

and the weight function is

$$(20) \quad w^{(\lambda)}(x; \varphi) = \pi^{-1} (2 \sin \varphi)^{2\lambda-1} e^{-(\pi-2\varphi)x} |\Gamma(\lambda+ix)|^2.$$

These polynomials may also be expressed in terms of hypergeometric series in the form

$$(21) \quad n! P_n^\lambda(x; \varphi) = (2\lambda)_n e^{inx} {}_2F_1(-n, \lambda+ix; 2\lambda; 1 - e^{-2i\varphi}).$$

These polynomials were mentioned by Meixner (1934) and by W. Hahn (1949). They have a representation in terms of finite differences, an analogue of Rodrigues' formula (Toscano, 1949). Setting

$$\delta F(x) = F(x + \tfrac{1}{2}i) - F(x - \tfrac{1}{2}i)$$

$$\delta^k F(x) = \delta[\delta^{k-1} F(x)] \quad k = 2, 3, \dots,$$

we have

$$(22) \quad P_n^\lambda(x; \varphi) = \frac{(-1)^n}{n!} \frac{\delta^n G(\lambda + \tfrac{1}{2}n, x)}{G(\lambda, x)}$$

where

$$G(\lambda, x) = \frac{\Gamma(\lambda+ix)}{\Gamma(1-\lambda+ix)} e^{2\varphi x}.$$

10.22. Orthogonal polynomials of a discrete variable

In the remaining sections of this chapter we shall briefly list a few systems of orthogonal polynomials for which the distribution function $\alpha(x)$ of sec. 10.1 is a jump function, and the appropriate definition of the scalar product is 10.1(3). The points at which the jumps of $\alpha(x)$ occur are x_i , and we shall use the *jump function* $j(x)$, the jump of $\alpha(x)$ at $x = x_i$ being $j(x_i)$. Thus, the appropriate definition of the scalar product is

$$(1) \quad (\varphi_1, \varphi_2) = \sum_i j(x_i) \varphi_1(x_i) \varphi_2(x_i),$$

and the jump function corresponds in some measure to the weight function of the earlier sections. We always assume that the jump function is positive and that $\sum_i j(x_i)$ is finite. Many results of the introductory sections of this chapter hold for scalar products of the form 10.1(2) and hence remain valid for the definition (1) of a scalar product.

The x_i will be taken as integers, $a \leq x_i \leq b$. The intervals and jump functions listed in the table below are those of most frequent occurrence. The orthogonal polynomials associated with them correspond to the classical orthogonal polynomials of a discrete variable, and most of them have been studied in some detail.

POLYNOMIALS OF A DISCRETE VARIABLE

a	b	$J(x)$	NAME
0	$N-1$	1	Tchebichef
0	N	$p^x q^{N-x} \binom{N}{x}$	Krawtchouk
0	∞	$\frac{e^{-a} a^x}{\Gamma(x+1)}$	Charlier
0	∞	$c^x \frac{(\beta)_x}{x!}$	Meixner
0	∞	$\frac{(\beta)_x (\gamma)_x}{x! (\delta)_x}$	W. Hahn

All these polynomials have a number of properties in common, among which we mention only the finite difference analogue of Rodrigues' formula

$$(2) \quad p_n(x) = [K_n j(x)]^{-1} \Delta^n [j(x-n) X(x) X(x-1) \cdots X(x-n+1)]$$

where K_n is a constant, $X(x)$ is a polynomial in x whose coefficients are independent of n , and Δ is the operator of forward differences,

$$(3) \quad \Delta f(x) = f(x+1) - f(x), \quad \Delta^{n+1} f(x) = \Delta[\Delta^n f(x)] \quad n = 1, 2, \dots$$

Conversely, this property characterizes the above orthogonal polynomials in the sense that any system of orthogonal polynomials possessing a Rodrigues' formula can be reduced to one of the systems listed above (Hahn 1949, Weber and Erdélyi 1952). The proof is analogous to that given in sec. 10.6 and will be omitted.

The proof of the orthogonal property of these polynomials may be based on (2) and "summation by parts". Alternatively, the method of generating functions may be used.

10.23. Tchebichef's polynomials of a discrete variable and their generalizations

Tchebichef's polynomials $t_n(x)$ arise in the graduation (fitting) of data by least squares. For an account of their properties see Szegő (1939, sec. 2.8), Jordan (1921 and 1947, Chapter VIII), and the references given in these places.

Definition and orthogonal property.

$$(1) \quad t_n(x) = n! \Delta^n \left[\binom{x}{n} \binom{x-N}{n} \right] \quad n = 0, 1, \dots, N-1$$

$$(2) \quad \sum_{x=0}^{N-1} t_m(x) t_n(x) = (2n+1)^{-1} N(N^2-1^2)(N^2-2^2) \dots (N^2-n^2) \delta_{mn} \\ m, n = 0, 1, \dots, N-1.$$

Symmetry and "central values".

$$(3) \quad t_n(N-1-x) = (-1)^n t_n(x)$$

$$(4) \quad t_{2m}(\tfrac{1}{2}N - \tfrac{1}{2}) = (-1)^m (2m)! \binom{2m}{m} \binom{\tfrac{1}{2}N - \tfrac{1}{2} + m}{m}$$

$$t_{2m+1}(\tfrac{1}{2}N - \tfrac{1}{2}) = 0.$$

Difference equation

$$(5) \quad (x+2)(x-N+2) \Delta^2 t_n(x) + [2x - N + 3 - n(n+1)] \Delta t_n(x) \\ - n(n+1) t_n(x) = 0.$$

Recurrence formula

$$(6) \quad (n+1)t_{n+1}(x) - (2n+1)(2x-N+1)t_n(x) + n(N^2-n^2)t_{n-1}(x) = 0 \\ n = 1, 2, \dots$$

Connection with Legendre polynomials.

$$(7) \quad \lim_{N \rightarrow \infty} N^{-n} t_n(Nx) = P_n(2x-1).$$

A generalization of Tchebichef's polynomials may be obtained by the definition

$$(8) \quad p_n(x; \beta, \gamma, \delta) = \frac{1}{n!} \frac{x! (\delta)_x}{(\beta)_x (\gamma)_x} \Delta^n \left[\frac{(\beta)_x (\gamma)_x}{(x-n)! (\delta)_{x-n}} \right].$$

In particular,

$$(9) \quad p_n(x; 1, a+1, a+1) = \frac{1}{n!} \Delta^n \left[\binom{x}{n} \binom{x+a}{n} \right],$$

and in this form it is immediately seen that

$$(10) \quad p_n(x; 1, 1-N, 1-N) = t_n(x).$$

Certain polynomials introduced by Bateman (1933) are also particular cases of (8). The polynomials (8) were introduced by Hahn (1949). They belong to the jump function

$$(11) \quad j(x; \beta, \gamma, \delta) = \frac{(\beta)_x (\gamma)_x}{x! (\delta)_x}.$$

The explicit formula

$$(12) \quad p_n(x; \beta, \gamma, \delta) = \frac{(\beta)_n (\gamma)_n}{n!} {}_3F_2(-n, -x, \beta + \gamma - \delta + n; \beta, \gamma; 1)$$

and a recurrence relation were given by Weber and Erdélyi (1952).

There is a connection with Jacobi polynomials,

$$(13) \quad \lim_{\gamma \rightarrow \infty} \gamma^{-n} p_n(\gamma x; a+1, \gamma, \gamma - \beta) = \binom{n+a}{a} P_n^{(a, \beta)}(2x+1).$$

10.24. Krawtchouk's and related polynomials

The orthogonal polynomials associated with the binomial distribution in probability theory were introduced by Krawtchouk (1929). They were studied by Aitken and Gonin (1935), and an account of their properties is found in Szegő's book (1939, sec. 2.8.2).

We assume

$$(1) \quad p > 0, \quad q > 0, \quad p + q = 1, \quad N \text{ a positive integer.}$$

Definition, jump function, orthogonal property.

$$(2) \quad k_n(x) = \frac{(-1)^n x! (N-x)!}{n! p^x q^{N-x}} \Delta^n \left[\frac{p^x q^{N-x+n}}{(x-n)! (N-x)!} \right] \quad n = 0, 1, \dots, N.$$

$$(3) \quad j(x) = \binom{N}{x} p^x q^{N-x}$$

$$(4) \quad \sum_{x=0}^N j(x) k_n(x) k_m(x) = \binom{N}{n} p^n q^n \delta_{nm} \quad m, n = 0, 1, \dots, N.$$

Explicit representation, generating function.

$$(5) \quad k_n(x) = q^n \binom{x}{n} F(-n, x - N; x - n; -p/q)$$

$$(6) \quad \sum_{n=0}^N k_n(x) z^n = (1 + qz)^x (1 - pz)^{N-x}.$$

The explicit representation shows the connection with the Jacobi polynomials; the (limiting) relations with Hermite polynomials and with the Charlier polynomials are given in Szegő (1939, p. 35, 36).

The special case $p = q = \frac{1}{2}$ has been studied by Gram (1882) and Greenleaf (1932).

The polynomials

$$(7) \quad m_n(x; \beta, c) = \frac{x!}{(\beta)_x} c^{-x-n} \Delta^n \left[\frac{c^x (\beta)_x}{(x-n)!} \right]$$

were investigated by Meixner (1934), Gottlieb (1938, $\beta = 1$), and other authors. (See references in Hahn 1949, p. 32). They are generalizations of Krawtchouk's polynomials.

$$(8) \quad p^n m_n(x; -N, -p/q) = n! k_n(x).$$

Explicit representation, jump function, orthogonal property.

$$(9) \quad m_n(x; \beta, c) = (\beta + x)_n F(-n, -x; 1 - \beta - n - x; c^{-1}) \\ = (\beta)_n F(-n, -x; \beta; 1 - c^{-1})$$

$$(10) \quad j(x) = c^x (\beta)_x / x!$$

$$(11) \quad \sum_{x=0}^{\infty} j(x) m_n(x; \beta, c) m_l(x; \beta, c) = n! (\beta)_n c^{-n} (1 - c)^{-\beta} \delta_{nl} \\ \beta > 0, \quad 0 < c < 1.$$

Symmetry, generating function

$$(12) \quad (\beta)_x m_n(x; \beta, c) = (\beta)_n m_x(n; \beta, c)$$

$$(13) \quad \sum_{n=0}^{\infty} m_n(x; \beta, c) \frac{z^n}{n!} = \left(1 - \frac{z}{c}\right)^x (1 - z)^{-x-\beta} \quad |z| < \min(1, |c|).$$

The explicit representation (9) leads to the following connections with Jacobi, Laguerre and Charlier polynomials.

$$(14) \quad m_n(x; \beta, c) = n! P_n^{(\beta-1, -\beta-n-x)}\left(\frac{2}{c} - 1\right)$$

$$(15) \quad \lim_{c \rightarrow 1} m_n\left(\frac{cx}{c-1}; \beta, c\right) = n! L_n^{\beta-1}(x)$$

$$(16) \quad \lim_{\beta \rightarrow \infty} \left[\left(-\frac{\beta}{a}\right)^n m_n\left(x; \beta, \frac{a}{\beta}\right) \right] = n! L_n^{x-n}(a) = (-a)^n c_n(x; a).$$

A recurrence relation and a difference equation have been given by Meixner (1934).

10.25. Charlier's polynomials

The polynomials introduced by Charlier are the orthogonal polynomials associated with Poisson's distribution of rare events in probability theory. They have been investigated by several authors among whom we mention Meixner (1934, 1938) and Doetsch (1933). For an account of their properties see Szegő (1939, sec. 2.8.1) and Jordan (1947, sec. 148).

Jump function, definition, orthogonal property.

$$(1) \quad j(x) = e^{-a} a^x / x! \quad a > 0, \quad x = 0, 1, 2, \dots$$

$$(2) \quad c_n(x; a) = \frac{x!}{a^x} \Delta^n \left[\frac{a^{x-n}}{(x-n)!} \right]$$

$$(3) \quad \sum_{x=0}^{\infty} j(x) c_m(x; a) c_n(x; a) = a^{-n} n! \delta_{mn}.$$

Explicit representations, generating function.

$$(4) \quad c_n(x; a) = \sum_{r=0}^n (-1)^r \binom{n}{r} \binom{x}{r} \frac{r!}{a^r}$$

$$(5) \quad c_n(x; a) = \frac{x! (-a)^{-n}}{(x-n)!} \Phi(-n, x-n+1; a)$$

$$(6) \quad \sum_{n=0}^{\infty} c_n(x; a) \frac{z^n}{n!} = e^{-z} \left(1 - \frac{z}{a}\right)^x \quad |z| < a.$$

A bilinear generating function was given by Meixner (1938).

Symmetry, recurrence relation, difference equation.

$$(7) \quad c_n(x; a) = c_x(n; a)$$

$$(8) \quad ac_{n+1}(x; a) + (x - n - a) c_n(x; a) + nc_{n-1}(x; a) = 0$$

$$(9) \quad ac_n'(x+1; a) + (n - x - a) c_n(x; a) + xc_n(x-1; a) = 0.$$

From the explicit representation (5) follows the connection with Laguerre polynomials

$$(10) \quad c_n(x; a) = (-a)^{-n} n! L_n^{x-n}(a).$$

The connection with Meixner's polynomials has already been given in 10.24(16).

REFERENCES

- Achyeser, N., 1934: *Comm. Inst. Sci. Math. Méc. Univ. Kharkoff (Zapiski Inst. Mat. Mech.)* (4) 9, 3-8.
- Aitken, A.C. and H. T. Gonin, 1935: *Proc. Roy. Soc. Edinburgh* 55, 114-125.
- Bateman, Harry, 1904: *Messenger of Math.* 33, 182-188.
- Bateman, Harry, 1905: *Proc. London Math. Soc.* (2) 3, 111-123.
- Bateman, Harry, 1933: *Tohoku Math. J.* 37, 24-38.
- Beckenbach, E.F., Wladimir Seidel and Otto Szász, 1951: *Duke J.* 18, 1-10.
- Bernstein, S., 1930: *Comm. Soc. Math. Kharkoff* (4) 4, 79-93.
- Bernstein, S., 1932: *Comm. Soc. Math. Kharkoff* (4) 5, 59-60.
- Bochner, Salomon, 1929: *Math. Z.* 29, 730-736.
- Brafman, Fred, 1951: *Proc. Amer. Math. Soc.* 2, 942-949.
- Burchnall, J.L., 1951: *Proc. London Math. Soc.* (3) 1, 232-240.
- Burchnall, J.L., 1952: *Quart. J. Math. Oxford* (2) 3, 151-157.
- Caton, W.B. and Einar Hille, 1945: *Duke Math. J.* 12, 217-242.
- Charlier, C.V.L., 1931: *Application de la théorie des probabilités à l'astronomie*, Gauthier - Villars.
- Cooper, R., 1950: *Proc. Cambridge Philos. Soc.* 46, 549-554.
- Doetsch, Gustav, 1933: *Math. Ann.* 109, 257-266.
- Doetsch, Gustav, 1935: *Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Nat.* (6) 22, 300-324.
- Erdélyi, Arthur, 1936: *Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Nat.* (6) 24, 347-350.
- Erdélyi, Arthur, 1937: *Math. Z.* 42, 641-670.
- Erdélyi, Arthur, 1937a: *J. London Math. Soc.* 12, 56-57.
- Erdélyi, Arthur, 1938: *Akad. Wiss. Wien. S.-B. IIa* 147, 513-520.
- Forsythe, G. E., 1951: *Duke J.* 18, 361-371.
- Gatteschi, Luigi, 1949: *Boll. Un. Mat. Ital.* (3) 4, 240-250.
- Gatteschi, Luigi, 1949a: *Rend. Mat. e applicazioni Roma* (5) 8, 399-411.
- Geronimus, J., 1944: *Doklady Akad. Nauk. S.S.S.R. (N.S.)* 44, 355-359.
- Giuliotto, Virgilio, 1939: *Ist. Lombardo, Rend.* 72, 37-57.
- Gottlieb, M.J., 1938: *Amer. J. Math.* 60, 453-458.
- Gram, J.P., 1882: *J. Math.* 114, 41-73
- Greenleaf, H.E.H., 1932: *Ann. Math. Statistics* 3, 204-255.
- Greenwood, R.E. and J.J. Miller, 1948: *Bull. Amer. Math. Soc.* 54, 765-769.